# Estimating the Price Effects of the Blocked JetBlue-Spirit Merger: A BLP Demand Estimation and Merger Simulation

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**Abstract**

In January 2024, a federal court blocked JetBlue's proposed acquisition of Spirit Airlines. This paper builds an independent structural analysis of the central question: what would the merger have done to prices and consumers? Because no public dataset links individual travelers to their choices, the analysis uses a constructed market-level panel of 41 routes over 12 quarters, with three routes anchored to observed fare and fuel-cost information. A BLP random-coefficients demand model is estimated by two-step efficient GMM, and Bertrand-Nash pricing under common ownership is simulated. On the three anchor routes, the merger raises average prices by about 52 percent for Spirit and 18 percent for JetBlue, with a modeled annualized consumer welfare loss of about $64.1 million across those three routes. Because the data-generating process is known, the estimates can be checked against their true values. The spread of price sensitivity across consumers is recovered well, but heterogeneity in taste for ultra-low-cost carriers is not identified, and this propagates to other estimates, including a mean price sensitivity about 12 percent above its true value. Results are limited to the three anchor routes and to upward pricing pressure.

# 1 Executive Summary

In January 2024, a federal court blocked JetBlue Airways' proposed acquisition of Spirit Airlines, agreeing with the U.S. Department of Justice that the merger would eliminate Spirit as an independent, ultra-low-cost competitor and raise prices (United States District Court for the District of Massachusetts, 2024). This analysis carries an independent demand model through to the same question the government raised: what would the merger actually have done to prices and to consumers.

A constructed market-level panel, calibrated wherever possible to fare, fuel, and route information from the case record, feeds a random-coefficients demand model, estimated using 2-step efficient GMM. Because this panel's data-generating process is fully known, every estimated parameter can be checked directly against its true value, an unusual advantage no researcher working with real market data has. The spread of price sensitivity across consumers is recovered well. The estimated demand system then supports a Bertrand-Nash merger simulation: marginal costs are recovered from observed prices and elasticities, the equilibrium-solving procedure is independently validated, and the model solves for the prices JetBlue and Spirit would jointly choose under common ownership.

On the 3 real anchor routes with verified fare data,[1] JetBlue and Spirit's prices rise substantially under the merger, averaged across all 36 route-quarter observations in the panel: Spirit by 51.6 percent, JetBlue by 18.0 percent. Rival carriers also see smaller increases, an expected consequence of reduced competitive pressure. An independently derived screening formula (the standard Farrell-Shapiro upward pricing pressure measure) confirms the same direction and a comparable magnitude, without relying on the full equilibrium solve. Translating these price changes into a dollar welfare effect uses the standard compensating variation formula for this class of demand model. Under the model and constructed data used here, the estimated annualized consumer welfare loss across those same 3 anchor routes is approximately $64.1 million.

[1] Boston to Fort Lauderdale, Boston to Miami, and Los Angeles to Fort Lauderdale.

Beyond its specific findings, this analysis is built to demonstrate a complete applied workflow. It defines a real market from public case and government data, constructs the data responsibly where real data does not exist, and specifies a structural model appropriate to the question. It finds and discloses exactly where that model's identification falls short, and it carries the results that do hold up through to a counterfactual decision that matters.

These results come with limits. This analysis is not an empirical replication of the government's own analysis, which relied on data that is not publicly available. A parameter capturing taste for ultra-low-cost carriers specifically is not recovered, and this propagates to the mean price sensitivity, which is about 12 percent above its true value; we disclose this finding and diagnose it in detail in the sections that follow. The welfare figure is not extrapolated to the complaint's more than 150 presumptively anticompetitive routes; the 3 anchor routes were chosen for their real, verified data, not as a representative sample.

This analysis also models upward pricing pressure only, holding every product's characteristics fixed and assuming no merger-specific efficiencies. It does not capture the fleet retrofit and product-elimination mechanism separately alleged in the government's complaint. For this reason, the price and welfare figures above should be read as the modeled upward-pricing-pressure component of the merger's potential consumer harm, not as an estimate of its total effect; the omitted mechanism could generate additional harm, but its magnitude and interaction with the modeled pricing response are not quantified here. The Limitations section sets out this scope in full. A short postscript addresses Spirit's actual cessation of operations in 2026, a real event caused by financial distress unrelated to the blocked merger, not a realization of anything modeled here.

# 2 Background

On July 28, 2022, JetBlue Airways Corporation agreed to acquire Spirit Airlines, Inc. for $3.8 billion (United States District Court for the District of Massachusetts, 2024; United States Department of Justice, Office of Public Affairs, 2023b). On March 7, 2023, the Antitrust Division of the U.S.

Department of Justice, joined by the attorneys general of Massachusetts, New York, and the District of Columbia, sued to block the deal (United States Department of Justice, 2023; United States Department of Justice, Office of Public Affairs, 2023b). An amended complaint filed March 31, 2023 added California, Maryland, New Jersey, and North Carolina as co-plaintiffs, bringing the total to six states and the District of Columbia (United States Department of Justice, Office of Public Affairs, 2023a).

The government's theory of the case centered on Spirit's role as an ultra-low-cost carrier (ULCC). The complaint cites Spirit's own internal estimates of this effect: fares on a route fall by an average of 17 percent when Spirit enters, and rise by an average of 30 percent when it exits (United States Department of Justice, 2023). The complaint identified more than 150 routes it considered presumptively anticompetitive under the proposed merger, covering more than 30 million passengers and more than $6 billion in annual revenue (United States Department of Justice, 2023). The complaint also alleged that JetBlue planned to retrofit Spirit's aircraft to JetBlue's own cabin standard, removing 10 to 15 percent of seats per plane and increasing seat pitch from 28 to 32 inches (United States Department of Justice, 2023).

Trial was held in the fall of 2023. On January 16, 2024, the U.S. District Court for the District of Massachusetts issued its Findings of Fact and Conclusions of Law, ruled for the government, and barred the merger (United States District Court for the District of Massachusetts, 2024). The court rejected JetBlue and Spirit's failing firm and weakened competitor defenses, finding that Spirit's own executives had testified to a plan to return to profitability and that JetBlue was not the only available purchaser (United States District Court for the District of Massachusetts, 2024). The injunction was narrow: it blocked the specific deal structure the parties had agreed to in July 2022, and did not foreclose a differently structured transaction in the future (United States District Court for the District of Massachusetts, 2024).

The government's claim is an example of what antitrust economists call a unilateral effects claim: harm that follows from the merged firm's own unilateral pricing decision, not from any

coordination with other airlines. Before the merger, when JetBlue raises its price, some of its customers switch to Spirit, and that diversion is a real cost to JetBlue, since those sales are lost to a rival. After the merger, if those diverted customers still switch to Spirit, a product the combined firm now also owns, the diversion is no longer entirely lost to the merged firm. The merger internalizes some of that diverted business, weakening JetBlue's incentive to keep its price low specifically on routes where Spirit is a close substitute.

The 2023 Merger Guidelines, issued jointly by the Department of Justice and the Federal Trade Commission, address this kind of harm directly in Guideline 2, which covers mergers that eliminate substantial competition between the merging firms (United States Department of Justice and Federal Trade Commission, 2023). The standard analytical tool economists use to evaluate this kind of harm is the diversion ratio: the share of a product's lost sales that flow to a specific rival product when that product's price rises. Estimating diversion ratios in a rigorous way generally requires a demand model that lets products substitute with each other in a realistic, product-specific way, rather than assuming every customer who leaves one product is equally likely to switch to every other product on the market. One widely used tool for this purpose is the random-coefficients discrete choice demand model developed by Berry, Levinsohn, and Pakes, commonly referred to as BLP (Berry et al., 1995), though it is far from the only model available for this kind of question: nested logit specifications, other discrete-choice models, and industry-specific structural models are also used in practice. This analysis uses a BLP-style specification because it directly addresses the substitution-pattern problem described above.

This analysis builds a demand model of this kind for the airline routes named in the government's complaint, and uses it to ask the same underlying question the government raised: what would the JetBlue-Spirit merger do to prices. It does not attempt to reproduce the government's own analysis, which relied on data that is not publicly available. It instead builds an independent model, calibrated to public information, and uses it to estimate the kind of unilateral effect the government's theory of harm describes.

# 3 Data and the Data-Generating Process

## 3.1 Overview

No public dataset links individual passengers to their itineraries, fares, and choices, the microdata a demand estimation of this kind would ideally use. The closest public source is the Airline Origin and Destination Survey, in which BTS publishes ticket-level samples of itineraries, fares, and passenger counts (DB1B, a 10 percent sample of tickets through June 2025, and its 40 percent successor DB1C) but no information on who bought them (Bureau of Transportation Statistics, 2025b). This analysis does not estimate on those data. Its purpose is to carry a BLP demand model through to a merger counterfactual on data where the truth is known, so the estimates can be checked, and no real dataset can supply known true parameters. The analysis therefore builds a constructed market-level panel. The panel draws on several real, publicly available sources, and where real data does not exist, supplements them with a purpose-built data-generating process; the result is a demand system whose true parameters are known by construction.

Throughout this section, and in Technical Appendix A, inputs are classified by source. A **real** input is taken directly from a public source, unchanged. A **real-derived** input is calculated from real data but requires an additional step, such as gap-filling or scaling. A **constructed** input is assigned or generated because no real data exists for it. **Calibration** is identified separately from these three: it describes a constructed value that is deliberately set to match a documented real-world target, rather than a fourth, mutually exclusive category alongside the other three.

The panel covers 41 routes over 12 quarters (2022 Q1 through 2024 Q4).[2] Each route-quarter is treated as a separate market: a self-contained group of consumers choosing among that route-quarter's available products, or the outside option of not flying, with no substitution assumed across

[2]The panel size of 41 routes (3 real anchor routes plus 38 constructed routes) was chosen from a pre-specified 30-to-50-route target range: large enough to give the GMM estimator meaningful cross-sectional variation, and small enough that each constructed route's characteristics could still be assigned and checked individually rather than mass-generated. There is nothing significant about 41 specifically; a panel of 40 or 42 routes, built the same way, would have served equally well.

markets. Between 2 and 4 competing products are present in a given market, depending on whether JetBlue and Spirit are both present, only one of them is present, or neither is.[3] This gives 492 route-quarter markets in total (41 routes times 12 quarters).

## 3.2 Real Data

### 3.2.1 Three Anchor Routes

Boston to Fort Lauderdale (BOS-FLL), Boston to Miami (BOS-MIA), and Los Angeles to Fort Lauderdale (LAX-FLL) were chosen because they are specifically named in the case record (United States Department of Justice, 2023; United States District Court for the District of Massachusetts, 2024). For these 3 routes, real JetBlue and Spirit fares were pulled from the Bureau of Transportation Statistics Consumer Airfare Report, which publishes quarterly average fares and market shares by carrier for domestic airport-pair markets (Bureau of Transportation Statistics, 2025a). Table 1 shows the average fare for each carrier on each route over the full 2022 Q1-2024 Q4 window. Some quarters had gaps in the published data. These gaps were filled using disclosed methods: cross-route fare ratios, a within-route seasonal ratio, and linear interpolation. Technical Appendix A documents which specific quarters were filled and which method was used for each.

*Table 1: Real Anchor Route Fares (Average, 2022 Q1-2024 Q4)*

| Route | JetBlue Fare | Spirit Fare | Nonstop Distance |
|---|---|---|---|
| Boston-Fort Lauderdale | $207.36 | $129.59 | 1,283 miles |
| Boston-Miami | $199.68 | $118.20 | 1,283 miles |
| Los Angeles-Fort Lauderdale | $354.06 | $137.25 | 2,343 miles |

*Note: BTS Consumer Airfare Report lists 1,283 miles for both Boston routes.*

[3] A legacy carrier and a Frontier-type ULCC are present in every market regardless of JetBlue and Spirit's presence, so every market has at least these 2 rivals. The 2 ULCC-type products, Spirit and the Frontier-type carrier, are included together specifically so the model can distinguish diversion to a similar ULCC product from diversion to a legacy carrier, which is central to the government's theory of harm and to the $\sigma_\gamma$ taste parameter discussed later in this analysis.

### 3.2.2 Real Fuel Costs

Quarterly jet fuel prices for the same window come from the U.S. Energy Information Administration's Gulf Coast kerosene-type jet fuel spot price series, retrieved through the Federal Reserve Bank of St. Louis (U.S. Energy Information Administration, 2025). These real fuel price movements are one of the 3 channels used to build price endogeneity into the data, described further below.

### 3.2.3 Real Route Composition

The Findings of Fact describe 562 routes the government identified as likely to be harmed by the merger.[4] These fall into 6 categories, each defined by a different competitive relationship between JetBlue and Spirit on that route:

- **Spirit-only routes (115):** Spirit is present and JetBlue is not (United States District Court for the District of Massachusetts, 2024, p. 44).
- **Spirit-entry routes (168):** routes Spirit planned to enter by 2027 but did not yet serve. JetBlue already served 23 of these as of November 2023 (United States District Court for the District of Massachusetts, 2024, pp. 44-45).
- **Econometric routes (96):** nonstop overlap routes where the government did not assert a structural presumption but anticipated harm through an econometric model (United States District Court for the District of Massachusetts, 2024, p. 45).
- **Connect routes (117):** both carriers compete only through connecting itineraries (United States District Court for the District of Massachusetts, 2024, pp. 45-46).
- **Mixed routes (15):** one carrier flies nonstop and the other connects (United States District Court for the District of Massachusetts, 2024, p. 46).
- **Nonstop overlap routes (51):** both carriers fly nonstop (United States District Court for the District of Massachusetts, 2024, p. 47).

[4]The 562-route total is the government's trial-stage figure from the Findings of Fact, spanning the 6 categories described below. It is a different measure from the complaint's earlier, pre-trial estimate of "more than 150" presumptively anticompetitive routes, described in the Background section above. At trial, the court treated the 183 routes in the nonstop overlap, mixed, and connect categories as carrying a structural presumption of anticompetitive effect; the remaining 379 routes did not (United States District Court for the District of Massachusetts, 2024, p. 77).

These 6 counts sum to the full 562.

The government built its 6 litigation categories to answer a specific legal question, how it proved harm on a given route. The demand model built here asks something narrower and more mechanical, simply which carriers are actually present in a market, since that determines which products exist and which pricing mechanism applies in the data-generating process. The government's 6 categories were regrouped into 4 categories that answer exactly this, and the regrouping is an exact partition of the real counts above, not an approximation:

- **both_jetblue_spirit**: nonstop overlap + mixed + connect + econometric routes (51 + 15 + 117 + 96 = 279 of 562, or 49.6%), because these 4 categories were all included in the government's analysis as involving competition or potential competition between JetBlue and Spirit, whether through direct nonstop competition, connecting competition, or the government's econometric assessment of competitive effects.
- **spirit_only**: the 115 Spirit-only routes (20.5%).
- **jetblue_only**: the 23 Spirit-entry routes where JetBlue already serves (4.1%).
- **neither**: the remaining 145 Spirit-entry routes, where neither carrier is present (25.8%).

These 4 category proportions, not the specific routes themselves, were used to assign the 38 constructed routes in this panel. The 38 constructed routes are not drawn from the government's list of 562 routes. They are constructed route pairs. Distance and market size for these routes are drawn from ranges that span above and below the 3 real anchor routes' own values, specified fully below under Constructed and Calibrated Data, consistent with the point-to-point network JetBlue and Spirit actually operate (United States District Court for the District of Massachusetts, 2024), rather than pulled from a real source. Each was assigned to one of the 4 categories only so that, in aggregate, the panel's category composition matches the real proportions above. The panel built here is therefore not a subset of the government's route list; only the category shares are real, calibrated to the government's own breakdown, while the specific routes making up 38 of the 41 markets are constructed.

Two qualifications apply to this route-composition data. First, the government's 562 routes are its own selected set of routes for its case, not a random sample of the U.S. air travel market. Second, the "neither" and "jetblue_only" categories both draw on Spirit's own forward-looking 2027 network expansion plan (the Spirit-entry routes) to distinguish which currently-Spirit-absent routes JetBlue already serves from which it does not, not a general control group of unrelated routes.

#### 3.2.4 Real Calibration Targets

Several figures cited in the government's case were used directly as calibration targets for the constructed parts of the data-generating process:

- Spirit's own estimate that fares fall by 17 percent on average when it enters a route (United States Department of Justice, 2023).
- JetBlue's own estimate that fares rise by 30 percent on average when Spirit exits (United States Department of Justice, 2023).
- The court's finding that JetBlue's own fare falls by more than 10 percent, and rival fares fall by 7 to 11 percent, when Spirit enters a market (United States District Court for the District of Massachusetts, 2024).

How these targets were used is described below.

## 3.3 Constructed and Calibrated Data

The remaining 38 routes have no real fare data at all. Their market size and distance were not pulled from any real source. Distance was drawn from a uniform distribution between 300 and 2,500 miles, and annual market size from a uniform distribution between 50,000 and 900,000 passengers, bracketing the 3 real anchor routes' own values (1,283 to 2,343 miles; 578,951 to 732,950 annual passengers) on both sides. The distance range is consistent with the point-to-point network JetBlue and Spirit actually operate: JetBlue's network centers on 6 focus cities (New York, Boston, the Miami/Fort Lauderdale area, Orlando, Los Angeles, and San Juan) (United States District Court

for the District of Massachusetts, 2024, pp. 15-16), and Spirit's on 3 core leisure markets (Fort Lauderdale, Orlando, and Las Vegas), with Los Angeles and New York City as growing additions (United States District Court for the District of Massachusetts, 2024, p. 21). None of these involve the shortest regional hops or the longest outlier routes, such as transpacific or Hawaii service, found elsewhere in the U.S. domestic network. Every carrier's fare on these routes, along with the legacy carrier and Frontier-type ULCC's fares on every route including the 3 real anchors, is constructed rather than observed, since no composite "legacy carrier" or "ULCC" actually exists in the real world data as a single, named entity.

## 3.4 The Demand System

Consumer $i$'s utility from product $j$ in a given market is

$$u_{ij} = \delta_j + \mu_{ij} + \varepsilon_{ij}$$

where $\delta_j$ is the mean utility of product $j$, common to every consumer in that market, $\mu_{ij}$ is consumer $i$'s individual deviation from that mean, and $\varepsilon_{ij}$ is an unobserved idiosyncratic shock. Mean utility itself is

$$\delta_j = X_j\beta - \alpha \cdot \text{price}_j + \xi_j$$

where $X_j$ contains observed product characteristics (seat pitch, nonstop status), $\beta$ their utility weights, and $\xi_j$ an unobserved product-market demand shock, the term at the center of the endogeneity discussion below. The individual deviation is

$$\mu_{ij} = -\sigma_\alpha \nu_i^\alpha \cdot \text{price}_j + \sigma_\gamma \nu_i^\gamma \cdot \text{ULCC}_j$$

where $\nu_i^\alpha$ and $\nu_i^\gamma$ are standard normal draws specific to consumer $i$, and $\sigma_\alpha$ and $\sigma_\gamma$ are the

standard deviations of price sensitivity and ULCC taste across the simulated population. Consumer $i$'s full price coefficient is therefore $\alpha + \sigma_\alpha \nu_i^\alpha$, with the negative sign in utility implying that higher prices reduce utility. Consumers also choose an outside option (not flying on this route at all), whose mean utility is normalized to 0.

## 3.5 Price Generation: Introducing Price Endogeneity

A meaningful test of the estimation strategy built later in this analysis requires prices to be correlated with unobserved demand, the type of correlation an instrumental variables strategy is designed to address. Without that correlation, there would be no genuine endogeneity problem for the instruments and the GMM estimator to correct, and their performance on this data would not speak to how they would perform on real, endogenous prices. Three real-world channels were built into the fare-generation process for this reason.

**Demand-side endogeneity:** Each product's fare responds directly to its own unobserved demand shock ($\xi_j$, from the mean utility decomposition above): a positive unobserved demand shock raises the constructed fare, the type of correlation that makes OLS estimation of a demand curve biased and motivates an instrumental variables approach.

**A real cost channel:** Fares respond to the real, quarter-by-quarter deviation in jet fuel prices described above. That fuel-price effect is scaled by nonstop distance, and the scaling factor itself is derived from the real, observed relationship between fuel price movements and fare movements on the 3 anchor routes.

**A competitive structure channel:** Fares include an adjustment that activates specifically when Spirit is absent from a market, sized directly to the real target figures described above (10 percent for JetBlue, 9 percent for rivals). This is what allows the data to contain an explicit, checkable competitive effect: removing Spirit from a market should raise the remaining carriers' fares by approximately the real, documented amount, and the calibration check in Technical Appendix A confirms that it does.

## 3.6 True Parameters

Because the data-generating process is fully known, the true values of every demand parameter are available, something a real empirical researcher working with real market data never has. These true values are used later in this analysis purely as a benchmark: to check how closely the estimation procedure recovers them, and to diagnose exactly where and why it does not. Table 2 reports the true demand-side parameters.[5]

*Table 2: True Demand-Side Parameters*

| Parameter | True Value | Description |
|---|---|---|
| $\alpha$ (mean) | 0.008824 | Mean price sensitivity |
| $\alpha$ (std. dev.) | 0.004 | Spread of price sensitivity across consumers |
| $\gamma$ (mean) | 0.0 | Mean ULCC taste |
| $\gamma$ (std. dev.) | 1.0 | Spread of ULCC taste across consumers |
| $\beta_{\text{seat pitch}}$ | 0.10 | Utility weight on seat pitch |
| $\beta_{\text{nonstop}}$ | 0.50 | Utility weight on nonstop service |
| JetBlue intercept | 1.0003 | Fixed utility advantage specific to JetBlue |

The mean price sensitivity was anchored to a representative airline own-price elasticity of approximately -1.5, placing the resulting elasticities within the -1 to -2 range documented as

[5] $\alpha$'s mean and standard deviation are anchored to a representative airline own-price elasticity benchmark, discussed in the text below. $\beta_{\text{seat pitch}}$ and $\beta_{\text{nonstop}}$ are not tied to an external target: their signs are fixed by economic logic (more legroom and nonstop service cannot reduce utility), and their magnitudes were set large enough to have a meaningful effect on utility without dominating the price and demand-shock terms. $\gamma$'s mean of 0 and standard deviation of 1.0 are set to isolate a single, cleanly interpretable source of heterogeneity: no consumer systematically likes or dislikes ULCC products on average, but individual consumers vary widely in that taste, which is exactly the heterogeneity the estimation is designed to test whether it can recover. The JetBlue intercept's derivation is described in the text below.

reasonable for the U.S. airline industry (Berry and Jia, 2010; Brons et al., 2002).

The JetBlue-specific intercept was added as a deliberate specification adjustment, not a calibration in the formal sense. An early version of the data-generating process produced a JetBlue market share well below its observed average share on the 2 anchor routes where both carriers operate. This was interpreted as evidence of a missing carrier-specific utility term, one that price, seat pitch, and nonstop status alone cannot capture. The intercept was derived using the closed-form logit odds relationship between a utility shift and the resulting change in market share, then fixed rather than iteratively tuned, so that the real share target was not mechanically built into the simulated demand system.

# 4 Empirical Strategy

## 4.1 Why a Random-Coefficients Model

A plain logit demand model has a well known limitation called independence of irrelevant alternatives. If one product's price rises, a plain logit model predicts that every other product gains share in the same proportion, regardless of how similar or dissimilar that other product actually is to the one whose price moved. Applied to this setting, a plain logit model would predict that if Spirit raises its fare, JetBlue, the legacy carrier, and the Frontier-type ULCC all pick up an equal share of Spirit's departing customers. That is not a realistic description of how airline passengers actually substitute. A passenger who chose Spirit because it was the cheapest option is more likely to switch to another cheap, no-frills option than to a full-service carrier at a much higher price.

The random-coefficients logit model developed by Berry, Levinsohn, and Pakes, commonly called BLP, addresses this by letting consumers differ from each other in how they value price and product characteristics (Berry et al., 1995). Because some consumers are more price sensitive than others, and some have a stronger taste for the ULCC product category, the model allows substitution patterns to differ across products according to these dimensions, rather than imposing

the proportional substitution of a plain logit model.

## 4.2 Demand Specification

The demand system is described in full in the Data section. Mean utility is $\delta_j = X_j\beta - \alpha \cdot \text{price}_j + \xi_j$, and each consumer's individual deviation from that mean is $\mu_{ij} = -\sigma_\alpha \nu_i^\alpha \cdot \text{price}_j + \sigma_\gamma \nu_i^\gamma \cdot \text{ULCC}_j$. Price appears in both terms because the model separates the mean price effect from consumer-specific heterogeneity. Adding $\delta_j$ and $\mu_{ij}$ together, consumer $i$'s total utility from product $j$ is

$$u_{ij} = X_j\beta + \xi_j + \sigma_\gamma \nu_i^\gamma \cdot \text{ULCC}_j + \varepsilon_{ij} - (\alpha + \sigma_\alpha \nu_i^\alpha) \cdot \text{price}_j$$

so the price coefficient consumer $i$ actually experiences is $\alpha + \sigma_\alpha \nu_i^\alpha$: a mean sensitivity shared by everyone ($\alpha$), plus that consumer's own random deviation from the mean ($\sigma_\alpha \nu_i^\alpha$). This decomposition separates the average price sensitivity from the consumer-specific component, allowing the model to estimate the mean effect and the degree of heterogeneity separately. The mean component is solved for numerically alongside $\xi_j$, while the individual-deviation component is computed directly from the trial value of $\sigma_\alpha$, the spread of price sensitivity across consumers, and a fixed set of simulated draws at every step.

The model contains 2 sets of parameters that need to be estimated: the linear parameters ($\beta$, $\alpha$), and the nonlinear parameters that govern how much consumers differ from each other ($\sigma_\alpha$, $\sigma_\gamma$). These 2 sets of parameters are handled differently within the nested estimation procedure described below.

## 4.3 Price Endogeneity and the Instrument Strategy

Fares in this dataset are, by construction, correlated with $\xi_j$, the unobserved demand shock: a positive demand shock raises the constructed fare, exactly as it would in a real market where firms price in response to demand conditions they observe but a researcher does not. This correlation

biases ordinary least squares. When demand is unexpectedly strong, $\xi_j$ is high, and the fare-generation process responds by raising the price, so a high observed fare is partly a symptom of strong demand rather than a cause of weak demand. OLS cannot separate these two effects: some of the demand strength that actually comes from $\xi_j$ gets misattributed to price instead, making a price increase look less harmful to demand than it truly is, and pulling the estimated price coefficient toward zero. Correcting for this requires instruments: variables that shift price but have no direct effect on demand except through price.

Four instruments were used: the deviation of that quarter's real jet fuel price from its sample average, nonstop distance, the interaction of these two instruments, and a count of ULCC rivals present in the market. Each has a specified and independently checked channel into fare generation, described in the Data section. Distance's validity as an instrument holds within this constructed data-generating process, where distance is built into fare generation but is deliberately left out of consumer utility. This should not be read as a claim that distance would satisfy the exclusion restriction in an actual airline demand application, where travelers plausibly value distance directly, through trip purpose, willingness to fly farther, or the value of the outside option, and not only through its effect on price. 3 additional BLP-style instruments were constructed from sums, across each market's competing products, of nonstop status, seat pitch, and rival count, but were dropped after testing showed none of these ever actually enters the fare-generation process in this dataset, and including them distorted the search for the nonlinear parameters described below. Only the 4 instruments with a specified, checked channel into price were used in the final estimation. The full construction and relevance evidence for all 7 instruments, including the 3 dropped, is documented in Technical Appendix C.

## 4.4 Estimation: A Nested Fixed Point Algorithm

The linear and nonlinear parameters are estimated within the nested fixed point algorithm introduced by Berry (1994) and refined in Berry et al. (1995), following the estimation approach set out in

Nevo (2000).

**Inner loop:** For any trial value of the nonlinear parameters ($\sigma_\alpha$, $\sigma_\gamma$), mean utility $\delta_j$ is not directly observed. It is solved for market by market using a contraction mapping: starting from a guess, predicted shares are compared to the market's real observed shares, and $\delta_j$ is adjusted up or down until predicted shares match observed shares exactly. This inversion was accelerated using SQUAREM, a general-purpose acceleration method for slow-converging fixed point iterations (Varadhan and Roland, 2008), which produced roughly a 14-fold reduction in computation time on the full panel without changing the converged result, confirmed by checking that predicted shares from the accelerated and unaccelerated versions matched closely. The full derivation, its verification against an exactly-solvable test case, and the verified speedup are documented in Technical Appendix B.

**Outer loop:** Once $\delta_j$ is solved for at a given trial ($\sigma_\alpha$, $\sigma_\gamma$), the linear parameters can be estimated by IV/GMM, yielding the corresponding GMM objective (described below). A numerical search (Nelder-Mead) varies ($\sigma_\alpha$, $\sigma_\gamma$) to find the values that minimize this objective.

## 4.5 The GMM Estimator

The linear parameters are estimated by 2-step efficient generalized method of moments (GMM), using the moment condition that the unobserved demand shock should be uncorrelated with the instruments: $E[Z_j \xi_j] = 0$.[6] The GMM objective, over the $N$ product-market observations in the panel, is

$$Q(\theta) = \left( \frac{1}{N} \sum_j Z_j \xi_j(\theta) \right)' W \left( \frac{1}{N} \sum_j Z_j \xi_j(\theta) \right)$$

[6] $Z_j$ is the full vector of variables assumed uncorrelated with $\xi_j$ for product-market $j$: the observed product characteristics that enter demand directly ($X_j$, seat pitch and nonstop status, which serve as their own instruments since they are already assumed exogenous) together with the 4 instruments used in the estimation (the fuel price deviation, nonstop distance, their interaction, and the ULCC rival count). None of these 4 enters the demand equation itself; each is assumed to shift price without entering utility directly, which is what lets them supply identifying variation for price, the one variable in the demand system that is not treated as exogenous.

where $W$ is a weight matrix. The first step uses $W = (Z'Z/N)^{-1}$; the second step re-weights using the heteroskedasticity-robust covariance of the first step's residuals, the standard efficient 2-step GMM procedure (Hansen, 1982).

## 4.6 Identification

The linear parameters are identified by the standard instrumental variables logic, though the 2 sets of linear parameters are identified differently from each other. The observed product characteristics ($X_j$: seat pitch, nonstop status) are assumed uncorrelated with $\xi_j$ on their own, so their coefficients ($\beta$) are identified directly from their own natural variation across products and markets, the same way an exogenous regressor is identified in any linear regression. Price is different, since it is correlated with $\xi_j$ by construction: its coefficient ($\alpha$) can only be identified from variation in price that is not itself contaminated by $\xi_j$. The 4 instruments supply exactly that variation, each shifting price without directly affecting demand, so the price variation they induce identifies how demand actually responds to price.

The nonlinear parameters, $\sigma_\alpha$ and $\sigma_\gamma$, are identified differently again. Each governs how much consumers differ from each other, in price sensitivity or in ULCC taste, and that heterogeneity leaves a trace in the aggregate data: it shapes how much of a product's gained or lost share, following a price change, comes from its closest substitutes rather than being spread evenly across every alternative. The outer loop's search finds the values of $\sigma_\alpha$ and $\sigma_\gamma$ that make the demand system's predicted substitution pattern, conditional on the instruments above, match the substitution pattern implied by the observed share data as closely as the GMM objective can achieve. In practice, the available instruments provide much stronger information about heterogeneity in price sensitivity than about heterogeneity in ULCC taste, making identification of $\sigma_\alpha$ substantially stronger than identification of $\sigma_\gamma$. This difference in identification strength is examined directly in the Results section.

# 5 Estimation Results

## 5.1 Parameter Estimates

Because the data-generating process is fully known, every estimated parameter can be compared directly against its true value, something a real empirical researcher working with real market data never has available. Table 3 reports the final estimates, alongside standard errors, t-statistics, and p-values for the 5 linear coefficients (see the Empirical Strategy section for the scope of what these standard errors cover, and Technical Appendix D for the full derivation).

*Table 3: Estimated vs. True Parameters*

| Parameter | Estimate | True Value | Std. Error | t-stat | p-value |
|---|---|---|---|---|---|
| $\sigma_\alpha$ | 0.003813 | 0.004 | – | – | – |
| $\sigma_\gamma$ | 0.000003 | 1.0 | – | – | – |
| $\alpha$ (mean) | 0.009907 | 0.008824 | 0.000363 | 27.31 | <0.001 |
| $\beta_{\text{seat pitch}}$ | 0.117507 | 0.10 | 0.004571 | 25.71 | <0.001 |
| $\beta_{\text{nonstop}}$ | 0.268653 | 0.50 | 0.105526 | 2.55 | 0.011 |
| $\beta_{\text{ULCC}}$ | -0.191582 | 0.0 | 0.083368 | -2.30 | 0.022 |
| $\beta_{\text{JetBlue}}$ | 0.995465 | 1.0003 | 0.085315 | 11.67 | <0.001 |

The mean price sensitivity, $\alpha$, and its standard deviation across consumers, $\sigma_\alpha$, are both recovered close to their true values. $\sigma_\alpha$ is within 5 percent of its true value; $\alpha$ is about 12 percent above its true value, roughly 3 standard errors away, a deviation discussed further below. $\beta_{\text{seat pitch}}$, $\beta_{\text{JetBlue}}$, and $\alpha$ are estimated precisely, with t-statistics above 11. $\beta_{\text{nonstop}}$ and $\beta_{\text{ULCC}}$ are estimated less precisely. Both are statistically distinguishable from zero under the reported conditional variance calculation, though $\beta_{\text{ULCC}}$ should not be interpreted substantively given the concurrent failure to recover $\sigma_\gamma$, discussed below. $\beta_{\text{nonstop}}$'s larger standard error is consistent with it being identified from a narrow slice of the panel, since JetBlue, Spirit, and the Frontier-type ULCC are

nonstop by construction in this dataset, leaving only the legacy carrier's nonstop status to vary.

$\sigma_\gamma$, the standard deviation of ULCC taste across consumers, is not recovered: the estimate is effectively 0 against a true value of 1.0. The rest of this section diagnoses why, ruling out a coding error as the explanation.

## 5.2 Why Gamma Is Weakly Identified

Before concluding the nonlinear search itself was the problem, the linear estimation machinery was checked in isolation, holding $\sigma_\alpha$ and $\sigma_\gamma$ fixed at their true values (a check no real researcher could perform, since the true values are never known outside a fully specified data-generating process like this one). With the true nonlinear parameters supplied directly, the linear coefficients recovered are close to their true values ($\alpha$ = 0.00918 vs. true 0.008824; $\beta_{\text{JetBlue}}$ = 1.020 vs. true 1.0003), confirming the GMM machinery itself works correctly. The failure is confined to the nonlinear search for $\sigma_\gamma$; the estimator itself is not the problem.

The central evidence for this is empirical. A grid search over the GMM objective, documented in full in Technical Appendix E, shows that $\sigma_\alpha$ has a clear interior minimum at its true value across the tested range, while $\sigma_\gamma$ does not: the objective simply prefers $\sigma_\gamma$ to be as small as possible everywhere on the grid. That empirical pattern, and not a theoretical claim about which instruments are required for identification, is what establishes that $\sigma_\gamma$ is weakly identified in this dataset.

A likely explanation traces to the instrument strategy. All 4 instruments used in this estimation (fuel price deviation, distance, their interaction, and the ULCC rival count) shift price, and $\sigma_\alpha$, the spread of price sensitivity across consumers, is identified because they generate genuine price variation to identify against. Identifying a random coefficient on a fixed characteristic does not strictly require an instrument that shifts that characteristic independently of price; exogenous product characteristics can themselves contribute to identification through the substitution patterns they generate across markets. In this constructed dataset, though, the combination of aggregate market-share data, the available variation in the ULCC characteristic, and the available instrument

structure does not supply enough independent information to pin down $\sigma_\gamma$. This is consistent with, though not strictly explained by, an instrument set that shifts price but not ULCC taste independently of it. In applied BLP work, heterogeneity of this kind is typically identified using individual-level "micro moments," data linking specific types of consumers to specific choices, rather than aggregate market shares alone (Petrin, 2002; Berry et al., 2004). No individual-level data of this kind exists in this dataset.

This failure has a second, traceable fingerprint in Table 3: $\beta_{\text{ULCC}}$ is estimated at -0.19, negative and statistically distinguishable from zero, against a true value of 0. This should not be read as independent evidence that consumers value the ULCC category less. In the data-generating process, ULCC taste enters only through the random coefficient ($\sigma_\gamma$), so the true linear effect is exactly 0 by construction. The negative estimate traces back to the same identification failure described above, rather than reflecting a separate finding. For any trial value of $\sigma_\gamma$, including an incorrect one, the inner loop's contraction mapping still finds a $\delta_j$ that exactly matches observed shares. Which $\delta_j$ is needed to do so, however, depends on $\sigma_\gamma$ itself, so a wrong $\sigma_\gamma$ requires a different, compensating $\delta_j$ than the correct one would. That compensating shift is systematically related to whether a product is a ULCC, since that is precisely the dimension $\sigma_\gamma$ was supposed to capture, so when this distorted $\delta_j$ is passed into the linear GMM step, some of it is picked up as a spurious linear effect on $\beta_{\text{ULCC}}$. The nonlinear search's failure to find the true $\sigma_\gamma$ does not stay contained to $\sigma_\gamma$ alone; it leaks into the linear coefficient estimated alongside it.

$\alpha$'s own point estimate shows a related, smaller version of the same pattern. When the nonlinear parameters are fixed at their true values, the check described above, $\alpha$ is recovered at 0.00918, about 4 percent above its true value of 0.008824. In the full joint estimation, where $\sigma_\gamma$ must be searched for rather than supplied, $\alpha$ is recovered at 0.009907, about 12 percent above the true value. The bias roughly triples once $\sigma_\gamma$ must be jointly estimated instead of fixed at the truth.

This was checked directly, rather than inferred by analogy to $\beta_{\text{ULCC}}$. $\delta_j$ was solved twice for every product-market row: once at the final estimated ($\sigma_\alpha$, $\sigma_\gamma$), and once at the true ($\sigma_\alpha$, $\sigma_\gamma$).

The difference between the two is the compensating shift attributable to the wrong $\sigma_\gamma$, and that shift correlates with whether a product is a ULCC at $r = -0.55$ across all 1,560 product-market rows, in the same direction as price's own correlation with ULCC status ($r = -0.67$): ULCC products receive both a more negative compensating $\delta_j$ shift (mean $-0.304$, versus $-0.091$ for non-ULCC products) and a lower fare (mean \$111.78, versus \$262.07 for non-ULCC products). The compensating shift and price are themselves directly correlated ($r = 0.20$), confirming the leakage channel empirically. The same $\delta_j$ distortion responsible for the spurious $\beta_{\text{ULCC}}$ estimate is also correlated with price, through the shared ULCC channel, so some of it leaks into $\alpha$'s identification as well. This establishes that the channel exists; it does not establish the exact share of $\alpha$'s 12 percent deviation the channel explains, and this analysis has not decomposed $\alpha$'s bias further. Even under the oracle-conditional check, where $\sigma_\gamma$ is supplied correctly, $\alpha$ remains about 4 percent above its true value. This residual is plausibly tied to the finite-sample bias documented generally for 2-step efficient GMM estimators, a source this analysis has not separately verified.

With $\sigma_\gamma$ effectively 0, the estimated model still functions as a random-coefficients model, since $\sigma_\alpha$ remains active and recovered close to its true value, but it operates with only one working source of consumer heterogeneity instead of two. The practical consequence for substitution patterns is described next.

## 5.3 Elasticities

For 2 products $j$ and $k$ in the same market, the own-price elasticity of product $j$ and the cross-price elasticity of product $k$ with respect to product $j$'s price are

$$e_{jj} = \frac{\text{price}_j}{\text{share}_j} \cdot \frac{\partial \,\text{share}_j}{\partial \,\text{price}_j}, \qquad e_{kj} = \frac{\text{price}_j}{\text{share}_k} \cdot \frac{\partial \,\text{share}_k}{\partial \,\text{price}_j}$$

Own-price elasticities are negative (a price increase reduces a product's own share); cross-price elasticities are positive (a price increase in one product shifts some share to its substitutes). Table 4

shows the full elasticity matrix for one illustrative market, Boston to Fort Lauderdale in the first panel quarter.

*Table 4: Elasticity Matrix, BOS-FLL Quarter 1 (Rows: Affected Product's Share; Columns: Whose Price Changed)*

| | Legacy | Frontier-type ULCC | JetBlue | Spirit |
|---|---|---|---|---|
| Legacy | -1.571 | 0.260 | 0.624 | 0.220 |
| Frontier-type ULCC | 0.894 | -1.528 | 0.679 | 0.249 |
| JetBlue | 0.879 | 0.278 | -1.402 | 0.238 |
| Spirit | 0.905 | 0.299 | 0.696 | -1.346 |

All 4 diagonal entries are negative and all 12 off-diagonal entries are positive, exactly as theory requires. One pattern in this table is worth stating precisely, without overstating it. The magnitudes within any single column sit close together rather than clearly differentiated: for a 1 percent Spirit price increase, the rightmost column, legacy gains 0.220, the Frontier-type ULCC gains 0.249, and JetBlue gains 0.238, a spread of only about 0.03. In this illustrative market, Spirit is the single largest cross-price effect whenever legacy, Frontier-type, or JetBlue raises price, and when Spirit itself raises price, the Frontier-type ULCC edges out JetBlue for the largest effect. With $\sigma_\gamma$ estimated at essentially 0, this ranking cannot be credited to any recovered ULCC-taste heterogeneity, since the estimation found none. It is better attributed to the one mechanism that is active, $\sigma_\alpha$, working through the real price differences between products. Price sensitivity heterogeneity interacting with the observed fare differences across products is enough, on its own, to give the estimated substitution pattern some economically sensible ordering. That ordering should not be mistaken for taste heterogeneity surviving the estimation in some attenuated form; the estimation found none. The magnitudes remain close together, and closer to the uniform substitution a plain logit model would predict, known as the independence of irrelevant alternatives property (the same limitation discussed in the Empirical Strategy section), than the true parameters would produce.

## 5.4 Diversion Ratios

The diversion ratio from product $j$ to product $k$ is the share of product $j$'s lost customers, following a price increase, that a specific rival $k$ picks up:

$$D(j \to k) = \frac{\partial \,\text{share}_k / \partial \,\text{price}_j}{-\partial \,\text{share}_j / \partial \,\text{price}_j}$$

A single market-quarter is not a reliable basis for a headline result: any 1 quarter can be dominated by that quarter's unobserved demand shock. Table 5 reports diversion from Spirit averaged across all 12 quarters on each of the 3 real anchor routes, alongside the diversion ratios the true parameters would have produced, an oracle comparison not available to a real researcher.

*Table 5: Diversion From Spirit, Averaged Across 12 Quarters*

| Route | Destination | Estimated | True |
|---|---|---|---|
| BOS-FLL | Legacy | 16.7% | 15.2% |
| BOS-FLL | Frontier-type ULCC | 34.9% | 42.0% |
| BOS-FLL | JetBlue | 44.2% | 39.1% |
| BOS-FLL | Outside option | 4.2% | 3.7% |
| BOS-MIA | Legacy | 11.6% | 10.3% |
| BOS-MIA | Frontier-type ULCC | 36.1% | 42.4% |
| BOS-MIA | JetBlue | 48.1% | 43.5% |
| BOS-MIA | Outside option | 4.3% | 3.8% |
| LAX-FLL | Legacy | 6.5% | 5.9% |
| LAX-FLL | Frontier-type ULCC | 52.0% | 57.1% |
| LAX-FLL | JetBlue | 29.6% | 26.5% |
| LAX-FLL | Outside option | 11.9% | 10.6% |

The direction of the gap is consistent across all 3 routes: diversion to the Frontier-type ULCC is understated, and diversion to legacy, JetBlue, and the outside option is overstated. This is the concrete, quantified consequence of $\sigma_\gamma$ being weakly identified: the model understates how much Spirit and the Frontier-type ULCC, the 2 ULCC products, pull from each other specifically, and spreads that missing pull toward the other options instead.

## 5.5 Why the Diversion Winner Varies by Route

Table 5 shows a pattern that needs explaining: Frontier-type does not win diversion on every route, despite being designed to be Spirit's closest substitute along both price and taste in the data-generating process. On BOS-FLL and BOS-MIA, JetBlue captures more of Spirit's departing customers than Frontier-type does; only on LAX-FLL does Frontier-type win as expected.

The explanation is a tug-of-war between 2 forces built into the demand system, not an inconsistency. JetBlue carries a large, fixed utility advantage ($\beta_{\text{JetBlue}} \approx 1.0$, described in the Data section), which pulls consumers toward it on every route regardless of price. The Frontier-type ULCC's fare, by construction, stays within 1 to 3 percent of Spirit's on every route. Which force wins depends on how large JetBlue's price premium over Spirit is on a given route, since a larger premium makes the fixed advantage harder to overcome. Table 6 shows this directly.

*Table 6: JetBlue's Price Premium Over Spirit and the Diversion Winner*

| Route | JetBlue's premium over Spirit | Diversion winner |
|---|---|---|
| BOS-FLL | 60.0% | JetBlue |
| BOS-MIA | 68.9% | JetBlue |
| LAX-FLL | 158.0% | Frontier-type ULCC |

LAX-FLL's much larger JetBlue-Spirit price gap, a consequence of JetBlue's fare rising with distance and LAX-FLL being the longest of the 3 anchor routes, is large enough to overcome

JetBlue's fixed advantage, letting the cheaper, closer substitute win diversion there instead. This is a structural feature of the estimated demand system rather than noise, and it illustrates that the model's predictions depend on the interaction of several estimated and constructed parameters together, not any one parameter in isolation.

# 6 Spirit Exit and the JetBlue-Spirit Merger

The estimated demand system from the previous section is used here to answer two related but distinct counterfactual questions. First, what happens to the surviving carriers' pricing power when Spirit exits a market, holding ownership structure fixed. Second, what happens to prices when JetBlue and Spirit merge, so that a price increase by one carrier no longer represents a full loss of business to the other. Both counterfactuals reuse the same estimated demand parameters and the same elasticity machinery from the Estimation Results section; neither involves any new estimation.

## 6.1 Spirit Exit

For the 4 simulated routes that receive a Spirit exit event in panel quarter 7 (identified directly from the data as routes with `both_jetblue_spirit` in the market before that quarter and `jetblue_only` after), Table 7 compares the own-price elasticities of the 3 surviving carriers before and after the exit, averaged across all pre-exit and post-exit quarters on all 4 routes.

*Table 7: Own-Price Elasticities Before and After Spirit's Exit*

| Carrier | Pre-Exit | Post-Exit | Change |
|---|---|---|---|
| Legacy | -1.872 | -1.849 | +0.023 |
| Frontier-type ULCC | -0.752 | -0.607 | +0.145 |
| JetBlue | -1.195 | -1.056 | +0.139 |

Every own-price elasticity moves toward 0 after Spirit's exit. An own-price elasticity closer to

0 means that the carrier's demand is less responsive to its own price, so a given percentage price increase produces a smaller percentage reduction in its share. This is the direct definition of reduced competitive constraint, since a firm facing less elastic demand can sustain a higher markup, the relationship formalized in the standard Lerner Index (Lerner, 1934). Frontier-type shows the largest change (a roughly 19 percent reduction in elasticity magnitude), JetBlue a meaningful but smaller change (roughly 12 percent), and legacy essentially no change (roughly 1 percent).

Table 8 shows the same before-and-after comparison for the cross-price elasticities among the 3 survivors.

*Table 8: Cross-Price Elasticities Among the Survivors, Before and After Spirit's Exit*

| Pair (affected, with respect to price changed) | Pre-Exit | Post-Exit | Change |
|---|---|---|---|
| Legacy, w.r.t. Frontier-type | 0.198 | 0.275 | +0.077 |
| Legacy, w.r.t. JetBlue | 0.742 | 0.923 | +0.181 |
| Frontier-type, w.r.t. Legacy | 0.159 | 0.316 | +0.157 |
| Frontier-type, w.r.t. JetBlue | 0.752 | 0.964 | +0.212 |
| JetBlue, w.r.t. Legacy | 0.159 | 0.323 | +0.164 |
| JetBlue, w.r.t. Frontier-type | 0.201 | 0.301 | +0.099 |

Every cross-price elasticity among the 3 survivors rises after Spirit's exit. To make 1 row concrete: "JetBlue, w.r.t. Legacy" asks how much JetBlue's own share moves when legacy raises its price by 1 percent. Before Spirit's exit, a 1 percent increase in legacy's price raises JetBlue's share

by about 0.159 percent; after the exit, the corresponding increase is 0.323 percent, roughly double. The reason is straightforward: with Spirit gone, a customer who leaves legacy because of a price increase has 1 fewer competing option. More of that departing traffic lands on JetBlue specifically, and, by the same logic, more of it lands on the Frontier-type ULCC as well, which is why every entry in the table rises, not just this 1 row. With Spirit no longer available as an outlet for departing customers, substitution among the remaining 3 carriers concentrates more tightly on each other, the same underlying shift in competitive pressure seen from the other direction in Table 7.

In the estimated model, this ranking is driven primarily by price proximity to Spirit, because the weakly identified $\sigma_\gamma$ contributes essentially no estimated ULCC taste heterogeneity. The Frontier-type ULCC's price sits within about 3 percent of Spirit's on every anchor route, making it the closest price substitute among the 3 survivors; legacy, priced 83 to 169 percent above Spirit across the anchor routes, was never a close price substitute to begin with. This ranking result can therefore be reported with more confidence than the raw diversion-ratio magnitudes in the previous section, which are known to be affected by the $\sigma_\gamma$ limitation.

## 6.2 The JetBlue-Spirit Merger

The merger counterfactual asks a different question: holding every product's characteristics fixed exactly as observed, what prices would JetBlue and Spirit jointly choose if one firm owned both, compared to the prices each chooses independently today. This isolates upward pricing pressure (UPP), the effect of common ownership on pricing incentives, from any operational change such as a fleet retrofit; see the Limitations section for the scope of what this does and does not capture.

**Recovering costs:** Each product's marginal cost is recovered from its observed price and elasticity, under the standard assumption that observed prices already reflect single-product profit maximization: $c_j = p_j + s_j/(\partial s_j/\partial p_j)$, equivalent to the more commonly cited $c_j = p_j(1 + 1/e_{jj})$. On BOS-FLL, quarter 1, this recovers costs of \$90.89 (legacy), \$59.55 (Frontier-type), \$57.79 (JetBlue), and \$38.16 (Spirit), all positive and below the observed price, which is consistent with

the assumed single-product Bertrand pricing structure.

**Validation:** Before trusting any counterfactual, the pre-merger system (each product owned independently) was solved from scratch, starting from the real observed prices, and confirmed to reproduce those same prices to within 4.55e-13, essentially machine precision. This validates the cost-recovery and equilibrium-solving procedure independently of the merger question itself.

**The merger counterfactual:** With JetBlue and Spirit jointly owned, the first-order pricing conditions for every product are solved simultaneously: $s_j(p) + \sum_{k \in F(j)} (p_k - c_k)\, \partial s_k / \partial p_j = 0$, where $F(j)$ is the set of products commonly owned with $j$ (only $j$ itself, pre-merger; $j$ and its merger partner, post-merger). Table 9 reports, for each carrier, a single average price change across all 36 route-quarter observations in the panel (3 anchor routes times 12 quarters each, combined into one figure per carrier, not a separate figure per route or per quarter).

*Table 9: Average Price Change From the JetBlue-Spirit Merger (Across All 36 Route-Quarter Observations)*

| Carrier | Average Price Change |
|---|---|
| Spirit | +51.6% |
| JetBlue | +18.0% |
| Frontier-type ULCC | +8.5% |
| Legacy | +0.4% |

The 2 merging parties show by far the largest increases, consistent with the upward pricing pressure predicted by common ownership in the model. Spirit's percentage increase is substantially larger than JetBlue's, which connects directly to the diversion findings in the previous section: a large share of Spirit's departing customers land specifically on JetBlue, so Spirit internalizes a large share of its own losses once the merger occurs, creating a strong incentive to raise price. The non-merging rivals also see smaller, positive increases. This is the umbrella effect: once the merging

firms raise their prices, rival products become relatively more attractive, allowing rivals to raise their own prices as well.

A concrete example shows where these percentages come from and why Spirit's 51.6 percent figure is larger than it might first appear. On BOS-FLL, quarter 1, the single-market walkthrough above recovers a pre-merger Spirit price of $148.41 and a marginal cost of $38.16, a margin of $110.25. The full equilibrium re-solve finds Spirit's post-merger price rising by $52.36 in that specific market-quarter, a 35.3 percent increase; JetBlue's price rises by $21.11 in the same market-quarter, a 10.5 percent increase (these are the same 2 dollar figures reported in Table 10 below, as part of the UPP cross-check). Two things are worth keeping in mind when reading the 51.6 percent average in Table 9. First, Spirit's fare is the lowest of the 4 products on every anchor route, so a dollar increase that is modest in absolute terms is mechanically a larger percentage increase on Spirit's low base fare than an identical dollar increase would be on JetBlue's higher one; part of the gap between Spirit's and JetBlue's percentage figures is this arithmetic, not evidence of a proportionally larger real effect on Spirit specifically. Second, the 36 route-quarters behind the average vary considerably, reflecting the same quarter-to-quarter demand-shock noise discussed throughout this analysis: Spirit's own average increase ranges from 28.7 percent on LAX-FLL to 70.9 percent on BOS-MIA across the 3 routes, and individual quarters swing further still within a single route. The reported 51.6 percent is the average across all of this variation, not a figure any single market-quarter actually produces.

Every result behind Table 9 (72 equilibrium solves: 3 routes x 12 quarters x 2 ownership scenarios each) was independently audited by recomputing the first-order condition residuals at the solved prices, confirming genuine convergence in every case, with 0 failures. The full audit methodology is documented in Technical Appendix F.

## 6.3 Simple UPP Robustness Check

The standard Farrell-Shapiro upward pricing pressure formula, computed directly from pre-merger data with no equilibrium solving required, is $\text{UPP}_j = D(j \to k) \times \text{margin}_k$, a dollar value representing the sales a merging firm no longer really loses once its partner captures them, not itself a predicted price increase. It is a narrower, static approximation to the full simulation above: it does not capture the subsequent equilibrium responses of the merging firms and rivals. Table 10 compares these 2 approaches, the simple UPP formula and the full equilibrium simulation, for BOS-FLL quarter 1.

*Table 10: Simple UPP vs. the Full Equilibrium Simulation*

| Carrier | Diversion Ratio to Partner | Partner's Margin | Simple UPP ($) | Full Simulation Price Increase ($) |
|---|---|---|---|---|
| Spirit | 38.1% | $143.77 | $54.73 | $52.36 |
| JetBlue | 23.0% | $110.25 | $25.39 | $21.11 |

In this application, the simple UPP figure is larger than the resulting equilibrium price increase in both directions, by about 4.5 percent for Spirit and about 20 percent for JetBlue, providing a useful robustness check on the direction and magnitude of the full simulation.

## 6.4 Consumer Welfare

The price changes in Table 9 translate into a dollar welfare effect using the standard compensating variation formula for logit-family demand models (Small and Rosen, 1981). The calculation proceeds in 4 steps, building directly on the utility specification from the Data section.

**Step 1 - Expected utility in each scenario:** Before the idiosyncratic shock $\varepsilon_{ij}$ is realized, consumer $i$'s expected utility from the best available option in a market, including the outside option, has a closed form known as the log-sum or inclusive value:

$$V_i = \ln\left(1 + \sum_j \exp(u_{ij})\right), \qquad u_{ij} = X_j\beta + \xi_j + \sigma_\gamma \nu_i^\gamma \cdot \text{ULCC}_j - (\alpha + \sigma_\alpha \nu_i^\alpha) \cdot \text{price}_j$$

where the leading 1 is $\exp(0)$, the outside option's normalized utility. This is computed twice for each market-quarter: once using the pre-merger prices ($V_i^{\text{pre}}$), once using the post-merger prices from Table 9 ($V_i^{\text{post}}$), holding every other component of $u_{ij}$ (observed characteristics, $\xi_j$, and each consumer's own $\nu_i^\alpha$, $\nu_i^\gamma$) fixed, so that the only thing changing between the 2 calculations is price.

**Step 2 - Convert the change in utility to dollars:** Since price enters utility with coefficient $-(\alpha + \sigma_\alpha \nu_i^\alpha)$, one dollar of price is worth $(\alpha + \sigma_\alpha \nu_i^\alpha)$ utils to consumer $i$, the same individual price coefficient introduced in the Data section. Dividing the change in $V_i$ by this quantity converts it from utils into dollars, consumer $i$'s individual compensating variation (Small and Rosen, 1981):

$$CV_i = \frac{V_i^{\text{post}} - V_i^{\text{pre}}}{\alpha + \sigma_\alpha \nu_i^\alpha}$$

This formula requires each consumer's price coefficient, $\alpha + \sigma_\alpha \nu_i^\alpha$, to be positive; a negative value would invert the sign of that consumer's compensating variation and would not represent a coherent economic preference. As discussed in the Data section, the normal distribution used for $\alpha_i$ in the true data-generating process permits roughly 1.34 percent of draws to be negative. This was checked directly against the 100 fixed simulation draws and the actual estimated parameters ($\alpha$ = 0.009907, $\sigma_\alpha$ = 0.003813) used throughout this section: none of the 100 draws produce a negative price coefficient, the smallest is 0.0019. The welfare figures in Table 11 are therefore not affected by this issue in practice, though it remains a general limitation of the normal specification, disclosed further in the Limitations section.

**Step 3 - Average across the simulated population:** Since $\nu_i^\alpha$ and $\nu_i^\gamma$ vary across the simulated consumers, $CV_i$ is computed separately for each one and then averaged, giving the population's

average per-capita welfare change for that market-quarter:

$$\overline{CV} = \frac{1}{N_{\text{draws}}} \sum_i CV_i$$

**Step 4 - Scale to a total dollar figure:** The per-capita compensating variation from Step 3 is a dollar figure per potential traveler; turning it into a total effect for a market-quarter requires knowing how many potential travelers that market-quarter represents. Each anchor route's market size is real-derived and is carried through the panel as an annual figure. It is divided by 4 here, the same annual-to-quarterly conversion used elsewhere in the data-generating process, since the panel itself is quarterly. Each anchor route's market size, on both bases, is:

| Route | Annual Market Size | Quarterly Market Size |
|---|---|---|
| BOS-FLL | 714,031 | 178,508 |
| BOS-MIA | 732,950 | 183,238 |
| LAX-FLL | 578,951 | 144,738 |

Multiplying the per-capita compensating variation for a given market-quarter by that route's quarterly market size converts the per-capita figure into a total dollar effect for that market-quarter, which is then averaged across all 3 anchor routes and all 12 quarters each.

Table 11 reports the result on 3 different bases, chosen to answer 3 different questions a reader might reasonably ask: what this costs a typical traveler (per-capita), what this costs a typical market in a typical quarter (average total), and what the aggregate annual scale of the harm is across the 3 routes this analysis actually covers (annualized, summed).

*Table 11: Consumer Welfare Effect of the Merger*

| Measure | Value | Basis |
|---|---|---|
| Per-capita welfare change | -$31.58 | Flat average across all 36 route-quarter observations |
| Average total welfare change, per market-quarter | -$5,337,611 | Flat average across all 36 route-quarter observations |
| Annualized welfare loss across the 3 anchor routes | -$64,051,335 | Each route's own 12-quarter average, summed across the 3 routes, then annualized |

The first 2 rows describe a typical single market-quarter, an average across all 36 route-quarter observations combined into one figure. The third row answers a different question, the combined annual impact across the 3 real anchor routes: each route's 12 quarters are averaged separately, those 3 route-level averages are summed rather than averaged again, and the total is scaled by 4 quarters to annualize it. It is not a sum of 12 separately realized quarterly figures. Because the merger simulation holds product characteristics and the choice set fixed, incorporates no efficiencies, and produces higher prices for every product in every market-quarter, the resulting consumer welfare change is negative in every market-quarter. This figure covers only the 3 real anchor routes used throughout this analysis. The government's complaint identified more than 150 presumptively anticompetitive routes in total; the 3 anchor routes were selected for their real, verified fare data, not as a representative sample of that larger set, and this welfare estimate should not be extrapolated to the full route set without separate justification.

# 7 Limitations

The results in the preceding sections should be read alongside the limitations below. None of them were discovered after the fact; each was identified during construction or estimation and is documented in full, with the specific diagnostic evidence, in the technical appendices. They are

grouped here by type rather than by which section they affect, since several limitations touch more than one result.

## 7.1 Data Coverage

Only 3 of the 41 routes in the panel have real fare data (Boston-Fort Lauderdale, Boston-Miami, and Los Angeles-Fort Lauderdale). The remaining 38 routes are constructed, with only their category proportions, not the specific routes themselves, calibrated to the government's real 562-route breakdown. The 3 real anchor routes were selected because they are specifically named in the case record, rather than because they form a random or representative sample of the government's identified route set. Every result that specifically depends on the anchor routes (the diversion ratios, the merger price effects, the welfare figures) should be understood as applying to those 3 routes, not to the complaint's more than 150 presumptively anticompetitive routes. A real-data extension using the public Origin and Destination Survey ticket samples could relax this restriction and is the natural next step; it would not by itself supply the passenger-level data needed to identify $\sigma_\gamma$.

## 7.2 Weak Identification of ULCC Taste Heterogeneity

$\sigma_\gamma$, the spread of ULCC taste across consumers, is not identified in this dataset: the estimate is effectively 0 against a true value of 1.0. The central evidence is the grid search documented in Technical Appendix E, which shows no interior minimum for $\sigma_\gamma$ anywhere on the tested range; a likely contributing explanation, discussed at length in the Estimation Results section, is that the available instruments shift price but not ULCC taste independently of it. This has a measurable, quantified consequence for the diversion ratios specifically, verified directly against the true parameters as an oracle comparison. Diversion to the Frontier-type ULCC is understated by roughly 5 to 7 percentage points across the 3 anchor routes, while diversion to legacy, JetBlue, and the outside option is correspondingly overstated. The merger price effects and welfare figures in Section 6 are built from the same biased demand system, so they can reasonably be expected to inherit some distortion from

the same source. This analysis has not, however, run an equivalent oracle comparison, recomputing the merger simulation and welfare calculation under the true parameters, to quantify the direction or size of that distortion for those specific outputs. The claim of a quantified, directional bias therefore applies to the diversion ratios in Section 5, not to every downstream result. The Spirit-exit ranking and the merger's UPP ranking are the 2 results least affected by this limitation regardless, since both are driven primarily by price proximity, a channel carried by the separately identified $\sigma_\alpha$, rather than by $\sigma_\gamma$ directly.

A related, narrower identification limitation affects $\beta_{\text{nonstop}}$, which is identified from a much smaller slice of the panel than the other linear coefficients, since JetBlue, Spirit, and the Frontier-type ULCC are nonstop by construction and only the legacy carrier's nonstop status varies. This shows up as a visibly larger standard error for $\beta_{\text{nonstop}}$ relative to the other coefficients in Table 3, though it remains statistically significant.

## 7.3 The Normal Price Coefficient

The random coefficient on price, $\alpha + \sigma_\alpha \nu_i^\alpha$, is specified as normally distributed, following standard BLP practice, rather than log-normally distributed. A normal specification permits a small share of simulated consumers to draw an economically implausible negative price coefficient, someone who would prefer higher prices. In the true data-generating process (mean 0.008824, standard deviation 0.004), this affects approximately 1.34 percent of a 100,000-draw sample. For the specific welfare calculation reported in Table 11, this was checked directly: none of the 100 fixed simulation draws used throughout the estimation and merger simulation produce a negative price coefficient under the estimated parameters. The specification limitation itself remains general, however, and would need to be revisited if this framework were extended to a larger number of simulation draws or a different set of estimated parameters.

## 7.4 Scope of Statistical Inference

Standard errors, t-statistics, and p-values are reported only for the 5 linear coefficients ($\alpha$, $\beta_{\text{seat pitch}}$, $\beta_{\text{nonstop}}$, $\beta_{\text{ULCC}}$, $\beta_{\text{JetBlue}}$), computed using the standard 2-step efficient GMM asymptotic variance formula, conditional on the final nonlinear parameter estimates (the full derivation is in Technical Appendix D). No formal standard error is reported for $\sigma_\alpha$ or $\sigma_\gamma$ themselves. A fully joint standard error covering all 7 parameters together would require numerically differentiating the moment conditions with respect to the nonlinear parameters as well, since mean utility is only implicitly defined by the inner loop's contraction mapping rather than available in closed form. The repeated-run stability evidence discussed in the Estimation Results section, with $\sigma_\alpha$ agreeing closely across independent searches of the outer loop while $\sigma_\gamma$ does not, provides the available diagnostic evidence on the 2 nonlinear parameters' relative reliability.

## 7.5 Scope of the Merger Simulation

The merger simulation models upward pricing pressure only: JetBlue and Spirit's pricing incentives change because each internalizes the other's diverted sales, with every product characteristic (seat pitch, nonstop status, marginal cost) held completely fixed. No merger-specific cost efficiencies are modeled or passed through to consumers. This is the standard zero-efficiencies baseline used in real merger review (Farrell and Shapiro, 2010), not a claim that real mergers never generate cost savings.

More importantly, the retrofit and product-elimination mechanism described in the government's complaint, JetBlue's stated plan to reconfigure Spirit's cabins to JetBlue's own standard, removing the ultra-low-cost, no-frills product category entirely from these markets, is not modeled here at all. This is a separate, additional harm mechanism from upward pricing pressure, not a restatement of it, and there are 2 separate reasons to expect it would add to the modeled UPP effect instead of offsetting it. First, from the structure of the discrete-choice problem, removing a product from a consumer's choice set, holding every other price fixed, can only weakly reduce that consumer's welfare: a smaller feasible set cannot yield a higher maximum attainable utility.

Second, and more directly, the Spirit Exit results in Section 6 already show that removing Spirit's ULCC product specifically loosens competitive constraint on the surviving carriers (own-price elasticities move toward 0, cross-price elasticities among the survivors rise), a channel distinct from, and additional to, the ownership-internalization effect the merger simulation models. Taken together, these give a defensible basis for treating the price and welfare figures reported in this analysis as the modeled upward-pricing-pressure component of the merger's potential consumer harm, not an estimate of its total effect; the omitted retrofit and product-elimination mechanism could generate additional harm, but its magnitude and interaction with the modeled pricing response have not been quantified here. The Spirit-exit results are a rough, imperfect proxy for the competitive consequence of losing the ultra-low-cost product category, since they do not also layer the ownership-driven pricing change onto JetBlue's own product the way a real retrofit-plus-merger would.

### 7.6 No Individual-Level Data

Every parameter in this analysis, real or estimated, is identified from aggregate route-quarter market shares. No passenger-level data, whether linking individual travelers to their choices or describing their demographics, was available or used. This is the direct cause of the $\sigma_\gamma$ identification limitation above: the applied BLP literature typically identifies taste heterogeneity along a fixed characteristic using "micro moments," data linking specific types of consumers to specific choices (Petrin, 2002; Berry et al., 2004), a category of data this analysis does not have access to.

## 8 Postscript: Spirit's 2026 Cessation of Operations

This analysis was constructed using market information through the end of 2024 and treats Spirit's exit, in the Spirit Exit section, as a hypothetical counterfactual: a route where Spirit stops competing, holding every other carrier's ownership and product unchanged. Since this analysis was built, Spirit Airlines ceased all operations on May 2, 2026, following a second Chapter 11 bankruptcy filing (Case No. 25-11897, filed August 29, 2025, after an earlier case, Case No. 24-11988, filed

November 18, 2024 and emerged from in early 2025) (Spirit Aviation Holdings, Inc., 2026), a failed $500 million federal rescue negotiation, and a sharp rise in jet fuel prices associated with the 2026 Iran war (National Public Radio, 2026; CNN Business, 2026). This real-world exit was not caused by the JetBlue-Spirit merger, which had been blocked more than 2 years earlier, and should not be interpreted as a realization of either counterfactual modeled here.

The mechanism matters, and the 2 exits are not economically equivalent. The merger counterfactual in this analysis models an *acquisition-driven* removal of Spirit: ownership changes, and the acquiring firm internalizes the diverted sales its own price increases would otherwise send to Spirit. Spirit's actual 2026 exit was a *financial collapse*: the airline became unable to continue operating at all, for reasons unconnected to any change in competitive ownership. These are different economic events, even though both remove Spirit as a competitor from the same markets. The real exit therefore raises a related structural question, not the same one this analysis directly answers.

That said, in terms of market structure, the real exit shares the key feature of the Spirit Exit counterfactual: Spirit's product disappears while the surviving carriers' ownership structures remain unchanged. This creates a genuine opportunity for future empirical work. Once sufficient post-exit fare and market share data become available for real routes Spirit used to serve, the qualitative predictions from the Spirit Exit section, own-price elasticities moving toward 0 for the surviving carriers and cross-price elasticities among them rising, could be checked directly against what actually happened in the market, a test that this analysis's constructed data cannot itself provide.

# 9 Conclusion

This analysis set out to answer a single question with a real demand model: what would the JetBlue-Spirit merger, as the government challenged it, do to prices and consumer welfare. The answer came from building each piece of the argument in sequence, rather than asserting a conclusion directly.

A constructed market-level panel, calibrated wherever possible to real fare, fuel, and route data

from the case record, fed a random-coefficients demand system, estimated using two-step efficient GMM. That estimation recovered the spread of price sensitivity well, with the mean about 12 percent above its true value, but did not recover ULCC taste heterogeneity, a limitation this analysis diagnoses and quantifies in detail. The resulting elasticities and diversion ratios showed how JetBlue, Spirit, and their rivals substitute for each other. A Spirit-exit counterfactual showed what removing Spirit from a market does to the surviving carriers' pricing power, holding ownership fixed. A merger counterfactual then showed what changes when ownership itself changes. JetBlue and Spirit's prices rise because each internalizes the sales it would otherwise lose to the other, a result validated against both a machine-precision reproduction of observed prices and an independent screening formula. Those price changes were then translated into a dollar consumer welfare loss on the 3 routes where real data made that translation possible.

Each of these results was carried through to the next: the estimated demand parameters became the elasticities, the elasticities became the diversion ratios and the recovered marginal costs, and those became the merger's equilibrium prices and the welfare figures built on top of them. No step stood on its own.

The limitations documented in this analysis, including the narrow real-data footprint, the weak identification of ULCC taste heterogeneity, the specification and inference choices, and the scope of the merger simulation itself, are not an afterthought. They define exactly what the results above can and cannot support, and they were identified and quantified during the course of building this analysis, not added after the fact to qualify a finished result. Spirit's actual cessation of operations in 2026, discussed in the Postscript, is treated the same way: as a real event that raises a related question, not as validation of anything modeled here.

What this analysis demonstrates is not a novel econometric method. It is a complete application of one: defining a real market, constructing data responsibly where real data does not exist, specifying a structural model appropriate to the question, finding and disclosing where that model's identification genuinely falls short, and carrying the results that do hold up through to a

counterfactual decision that matters.

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# Technical Appendix A: Data Construction Detail

This appendix documents, in full, how the 3 real anchor routes' data was used both directly and as the source for several constants used throughout the construction of the other 38 routes. Every derivation below is computed by an explicit function from the real fare data, not hardcoded, so it is auditable and reproducible rather than asserted.

## A.1 Real Anchor Route Fare Data and Gap-Filling

Real JetBlue and Spirit fares for the 3 anchor routes come from the BTS Consumer Airfare Report (Bureau of Transportation Statistics, 2025a). A route's fare for a given carrier and quarter is missing from that report when a different carrier, rather than JetBlue or Spirit, was the lowest-fare carrier that quarter, since the report identifies the lowest-fare carrier by route rather than reporting every carrier's fare unconditionally. Each route had a different pattern of gaps, filled by a different, disclosed method appropriate to that gap:

- **BOS-FLL**: 1 gap, Spirit's 2022 Q1 fare (American was the lowest-fare carrier that quarter). Filled using BOS-FLL's own average Q1-to-Q2 seasonal ratio, computed from the years with real data for both quarters (Spirit's fare runs about 6.4 percent higher in Q1 than Q2), applied to the real 2022 Q2 value.
- **BOS-MIA**: every gap filled using the cross-route ratio to BOS-FLL for the same quarter and carrier, applied to BOS-FLL's own real (or, for 2022 Q1 Spirit, already seasonally estimated) value.
- **LAX-FLL**: 2 gaps, Spirit's 2024 Q3 and Q4 fares (Delta was the lowest-fare carrier both quarters). Filled by linear interpolation between the real 2024 Q2 fare and the real 2025 Q1 fare, the latter falling outside the panel's 2022-2024 window and used only as an external reference point for the interpolation, not as part of the simulated panel itself.

Every filled value is flagged with an `is_fare_estimated` indicator in the underlying data, so

gap-filled and directly observed values remain distinguishable throughout the panel.

## A.2 Deriving the JetBlue Fare-Versus-Distance Line

Every route's JetBlue reference fare, including the 38 constructed routes with no real data at all, is generated from a fare-versus-distance line: fare $=$ base $+$ (per_mile $\times$ distance). The 2 constants in this line were derived from real data, using only 2 of the 3 anchor routes: BOS-FLL (real average fare \$207.36 at 1,283 miles) and LAX-FLL (real average fare \$354.06 at 2,343 miles). BOS-MIA was deliberately excluded from this specific calculation, since JetBlue is not the largest-share carrier there and only 5 of its 12 quarters are real rather than gap-filled; including it would mean the "real" anchor point used to build every other route's fares is itself mostly estimated, defeating the purpose of anchoring to real data at all.

With exactly 2 real points, this is 2 equations and 2 unknowns, solved algebraically for an exact solution rather than a statistical fit. There is no regression, no residual, and no measure of fit with only 2 points; this line should be understood as an exact calibration through 2 real data points, not an estimated relationship between fare and distance. Solving:

$$\text{per_mile} = \frac{\$354.06 - \$207.36}{2{,}343 - 1{,}283} = 0.1384, \qquad \text{base} = \$207.36 - (1{,}283 \times 0.1384) = \$29.80$$

## A.3 Deriving Endogeneity Sensitivity Coefficients

The strength of the relationship between a product's fare and its own unobserved demand shock, $\xi_j$, discussed in the Data section, is not a single constant applied uniformly; it differs by carrier type, set by that carrier's own real fare volatility. For each of JetBlue and Spirit, the real fare coefficient of variation (CV, standard deviation divided by mean) is computed separately within each of the 3 anchor routes first, then averaged across the 3 routes, using only real, non-estimated fares. Computing the CV separately within each route before averaging, rather than pooling every

real fare from all 3 routes together, matters because the 3 routes have very different price levels (LAX-FLL averages around $354 for JetBlue versus $207 on BOS-FLL); pooling would conflate genuine quarter-to-quarter volatility with the difference in price levels between routes. This gives JetBlue a coefficient of 0.1046 and Spirit a coefficient of 0.1310.

Legacy and the Frontier-type ULCC have no real fare data at all, since neither is an actual named carrier, so each borrows the coefficient of the real carrier it is economically closest to: legacy, a differentiated rather than ultra-low-cost carrier, borrows JetBlue's 0.1046; the Frontier-type ULCC borrows Spirit's 0.1310.

This coefficient of variation is a real, measured statistic, and its use here is disclosed as the least-grounded assumption in the data-generating process: it measures real, observed fare volatility, and stands in for how strongly a fare might respond to an unobserved demand shock, but it is not itself a measurement of that sensitivity, since $\xi$ is a theoretical construct with no directly observable real-world counterpart.

## A.4 Deriving the Fuel Cost Scaling Factor

The real cost channel described in the Data section, fares responding to real quarterly jet fuel price deviations, uses a scaling factor derived from a different piece of the real fare data than Section A.2 above. Where the JetBlue fare line uses only each route's average real fare (a cross-route, between-route relationship, 2 numbers total), this calculation uses the leftover variation within each route over time: how much a given quarter's real JetBlue fare deviated from that route's own average, related to how much that quarter's real fuel price (U.S. Energy Information Administration, 2025) deviated from the sample window average, multiplied by that route's real distance. Using both pieces of real data does not double-count the same information, since one is a between-route relationship and the other a within-route, over-time relationship.

A single scaling factor is solved for via ordinary least squares through the origin (no intercept, since a fare deviation of exactly 0 is expected when the fuel price deviation is exactly 0), using all

29 real JetBlue fare observations across the 3 anchor routes:

$$\text{scaling_factor} = \frac{\sum(\text{fuel_deviation} \times \text{distance}) \times \text{fare_deviation}}{\sum(\text{fuel_deviation} \times \text{distance})^2} = 0.012402$$

This is a data-derived calculation from actual fare and fuel data, not an assumed or invented magnitude. It should still be understood as a simple, disclosed estimate from a small number of real observations (29), rather than a fully specified structural cost model.

## A.5 Seat Pitch

JetBlue's seat pitch (32 inches) and Spirit's and the Frontier-type ULCC's (28 inches) come directly from the government's complaint, which cites these exact figures in describing JetBlue's planned retrofit of Spirit's cabins (United States Department of Justice, 2023). Legacy's seat pitch (31 inches) is not derived from the anchor routes; it is a representative middle value for major legacy carriers such as American, Delta, and United, whose real seat pitch typically falls in the 30 to 31 inch range (AwardLocker, 2026).

## A.6 Constant Source Summary

*Table 12: Source of Key Constructed-Data-Generating-Process Constants*

| Constant | Value | Derived From | Source |
|---|---|---|---|
| JetBlue fare line (base) | $29.80 | BOS-FLL and LAX-FLL real average fares | Real-derived (exact, 2-point) |
| JetBlue fare line (per-mile) | 0.1384 | BOS-FLL and LAX-FLL real average fares | Real-derived (exact, 2-point) |

| Constant | Value | Derived From | Source |
|---|---|---|---|
| Endogeneity sensitivity, JetBlue/legacy | 0.1046 | Real JetBlue fare CV, all 3 anchor routes | Real-derived |
| Endogeneity sensitivity, Spirit/Frontier-type | 0.1310 | Real Spirit fare CV, all 3 anchor routes | Real-derived |
| Fuel cost scaling factor | 0.012402 | Real fare and fuel deviations, all 3 anchor routes | Real-derived (29 observations) |
| Seat pitch, JetBlue | 32 in. | Government's complaint | Real |
| Seat pitch, Spirit/Frontier-type | 28 in. | Government's complaint | Real |
| Seat pitch, legacy | 31 in. | Industry-typical range | Illustrative |
| Route category proportions | 49.6% / 20.5% / 4.1% / 25.8% | Findings of Fact, 562-route breakdown | Real |
| Spirit-exit fare adjustment (JetBlue) | +10% | Findings of Fact | Calibrated to real target |
| Spirit-exit fare adjustment (rivals) | +9% | Findings of Fact | Calibrated to real target |
| Mean price sensitivity, $\alpha$ | 0.008824 | Anchored to a representative airline elasticity of -1.5 | Calibrated to real target |

| Constant | Value | Derived From | Source |
|---|---|---|---|
| JetBlue-specific utility intercept | 1.0003 | Real JetBlue share on the 2 anchor routes where both carriers operate | Specification adjustment, see Data section |

## A.7 Calibration Check: Does the Simulated Spirit Exit Effect Match Reality

The competitive structure channel described in the Data section is validated by comparing the data-generating process's own simulated Spirit exit effect against the real target figures already introduced in the Background section. This comparison is averaged across 30 independent simulation runs, to smooth out the noise inherent in only 4 routes receiving a simulated exit event per run.

This is a sanity check on the data-generating process's assumed parameters, not the BLP estimation itself. It is also explicitly a mirror-image check for 3 of the 4 real targets below, since no entry events are simulated, only exits. A mirror check like this confirms only that the simulated effect points the right way and falls in a plausible range; it falls short of a strict test against a real exit-specific figure, since no such figure exists for those 3 targets.

*Table 13: Calibration Check, 30-Run Average*

| Check | Real Target | Simulated (Mean) | Simulated (Std. Dev.) |
|---|---|---|---|
| Industry-wide average fare, direct check against JetBlue's estimate of the industry-wide effect | 30% | 21.8% | 5.2% |

| Check | Real Target | Simulated (Mean) | Simulated (Std. Dev.) |
|---|---|---|---|
| Industry-wide average fare, mirror of Spirit's estimate of the industry-wide effect | 17% | 21.8% | 5.2% |
| JetBlue's own fare, mirror of the Findings of Fact's entry estimate | 10% | 9.4% | 8.4% |
| Rival fares, mirror of the Findings of Fact's entry estimate | 7-11% | 14.3% | 9.5% |

"Industry-wide average fare" above means the average fare across every carrier active in the market, not any one carrier's fare specifically; it is distinguished from "JetBlue's own fare" and "Rival fares," which are carrier-specific averages covering only JetBlue or only the 2 surviving rivals (legacy and the Frontier-type ULCC), respectively.

The simulated effects fall in the same general range as the real targets, with sizeable standard deviations reflecting the genuine small-sample noise from only 4 exiting routes per run, the same reason the 12-quarter and 3-route averaging discipline is used throughout the main results rather than reporting any single run or market in isolation. This validates that the competitive structure mechanism in the data-generating process produces an effect of roughly the right size and direction, not that it reproduces the real figures precisely; with only 4 simulated exit events available for calibration, precise reproduction was never the goal.

# Technical Appendix B: SQUAREM Acceleration

This appendix documents the SQUAREM-accelerated version of the inner loop's contraction mapping: its derivation, the numerical safeguard that governs when an accelerated step is accepted, and the verified speedup and converged-solution agreement against the plain, unaccelerated version.

## B.1 The Baseline Contraction Mapping

For a given trial value of the nonlinear parameters, mean utility $\delta_j$ is recovered market by market using Berry's (1994) contraction mapping: starting from a guess, predicted shares are compared to the market's real observed shares, and each $\delta_j$ is adjusted up or down until predicted shares match observed shares exactly. Each ordinary step closes some fraction of the remaining gap to the true answer; when that fraction is small, convergence can take many steps even though every step moves in the right direction.

## B.2 SQUAREM: Mechanics, Derivation, and Verification

SQUAREM (Varadhan and Roland, 2008) accelerates this without changing what it converges to. In each cycle, given a current $\delta$:

1. Take 1 ordinary contraction step to get $\delta_1$.
2. Take 1 more ordinary step from $\delta_1$ to get $\delta_2$.
3. Compute $r = \delta_1 - \delta$ (the first step taken) and $v = (\delta_2 - \delta_1) - r$ (how the step changed on the second try).
4. Compute the extrapolation length, $\alpha = (r \cdot r)/(r \cdot v)$, the dot product of $r$ with itself over the dot product of $r$ with $v$, the standard "S1" scheme from Varadhan and Roland (2008). Because $r$ and $v$ are vectors once a market has more than 1 product, this dot-product form correctly accounts for the direction, not only the size, of how the step changed between the 2 ordinary iterations.

5. Compute a single extrapolated candidate, $\delta_{\text{candidate}} = \delta - 2\alpha r + \alpha^2 v$, typically landing far closer to the fixed point than the 2 ordinary steps alone.
6. Compare how close the candidate's implied shares come to the observed shares against how close the safe, ordinary 2-step result $\delta_2$ comes, and accept the candidate only when it is genuinely at least as good under that comparison; otherwise fall back to $\delta_2$.

This safeguard ensures that the extrapolated candidate is accepted only when it is at least as good as the safe, ordinary 2-step result under the convergence criterion, while typically requiring far fewer cycles to converge.

**Verification against an exactly-solvable case:** Before relying on this extrapolation on real market data, the formula in Step 4 was checked against a simple linear fixed-point map with a known, closed-form answer: $x_{\text{new}} = 0.5x + 3$, whose exact fixed point is $x^* = 6$. Starting from $x_0 = 0$: $x_1 = 3$, $x_2 = 4.5$, so $r = 3$ and $v = (4.5-3)-3 = -1.5$. Then $\alpha = (3\times 3)/(3\times -1.5) = -2$, and $x_{\text{candidate}} = 0 - 2(-2)(3) + (-2)^2(-1.5) = 12 - 6 = 6$, exactly the true fixed point, in a single cycle from a starting guess that had not moved at all.

### B.3 Verified Speedup and Converged-Solution Agreement

On the full panel (492 route-quarter markets), the SQUAREM-accelerated inversion ran substantially faster than the plain, unaccelerated contraction mapping without changing the converged $\delta$ values. The original run showed roughly a 14-fold speedup. An independent re-run, performed to confirm this generalizes across environments, showed an 18.94-fold speedup (97.16 versus 5.13 seconds), with the 2 versions' final $\delta$ values agreeing to within $1.00 \times 10^{-7}$. The exact factor is hardware-dependent, as expected; what matters is that SQUAREM changes speed, not the converged solution.

## Technical Appendix C: Instrument Construction and Relevance

This appendix documents the full instrument construction, all 7 instruments originally built following standard practice, and the evidence behind keeping 4 of them and dropping the other 3.

## C.1 The 7 Constructed Instruments

Two instrument families were built, drawing on 2 established approaches to instrument construction in differentiated-product demand estimation.

**Cost-shifter instruments (3)**: the deviation of that quarter's real jet fuel price from its sample-window average, nonstop distance, and the interaction of the 2. These are motivated by the standard logic that real, external cost variation shifts price without directly affecting demand.

**BLP-style rival-characteristic instruments (4)**: for each product, the number of rival products in the same market, the sum of rivals' seat pitch, the sum of rivals' nonstop status, and the number of rival ULCC-type products specifically (Spirit or the Frontier-type ULCC). These are motivated by the standard logic, common in the applied BLP literature, that a firm's markup in a full equilibrium pricing model responds to the characteristics of the products it competes against, not only its own characteristics.

## C.2 The Relevance Test: a Direct Code-Level Check, Not a Correlation Ranking

Because this analysis's data-generating process is fully known, each instrument's relevance could be checked directly, by examining whether that variable is actually used as an input anywhere in the fare-generation code, rather than inferred indirectly the way a real researcher working with real market data would have to. This direct check found that distance and the fuel price deviation are used as inputs to every simulated fare, and that the rival ULCC count is used as a direct input to Spirit's and the Frontier-type ULCC's own fare, through a competitive discount mechanism tied specifically to the ULCC dimension. The remaining 3 instruments, the number of rivals generally, the sum of rivals' seat pitch, and the sum of rivals' nonstop status, are never used as inputs anywhere in the fare-generation code; only the count of rival ULCC-type products specifically has a real channel into price, not rival characteristics generally.

Table 14 reports each instrument's raw correlation with fare, computed directly from the panel, alongside whether it was kept for estimation. This table is included as supporting, illustrative evidence, not as the basis for the keep-or-drop decision itself, since the direct code-level check in the previous paragraph already establishes whether an instrument has a channel into price in this constructed DGP. The correlation table alone is not a clean predictor of that channel: `instrument_fuel_deviation`, one of the 4 kept instruments, has a smaller raw correlation with fare (0.0711) than 2 of the 3 dropped instruments (`instrument_n_rivals` at -0.1052, `instrument_rival_seat_pitch_sum` at -0.1440). What determines relevance here is a verified causal channel into price, not the size of a raw correlation.

*Table 14: Instrument Relevance*

| Instrument | Correlation with Fare | Kept for Estimation |
|---|---|---|
| Fuel price deviation | 0.0711 | Yes |
| Distance | 0.6388 | Yes |
| Fuel x distance interaction | 0.0818 | Yes |
| Rival ULCC count | 0.3850 | Yes |
| Number of rivals | -0.1052 | No |
| Sum of rivals' seat pitch | -0.1440 | No |
| Sum of rivals' nonstop status | -0.0110 | No |

## C.3 Why the 3 Irrelevant Instruments Were Dropped, Not Merely Left Unused

Including the 3 instruments without a verified causal channel was found, empirically, to distort the outer loop's search for the nonlinear parameters: with them included, the GMM objective moved monotonically away from the known true parameter values rather than toward them, providing direct evidence that including the additional instruments was distorting the estimation in this constructed

setting. This distortion could only be diagnosed directly because the true parameters are known here, a check that would not be available to a real researcher working with real market data, who would instead need to assess relevance using first-stage diagnostics and other identification checks. Only the 4 instruments with a verified, direct channel into price, the 3 cost shifters and the rival ULCC count, were used in the final estimation reported in the Estimation Results section.

### C.4 A Second Instrument Family, Added Mid-Construction

The rival ULCC count instrument was not part of the original instrument set. It was added after an earlier estimation attempt using only the 3 cost-shifter instruments. That attempt found the 3 cost shifters could identify the price sensitivity random coefficient ($\sigma_\alpha$) but had essentially no power to identify the ULCC taste random coefficient ($\sigma_\gamma$). The reason is structural: cost shifters are naturally suited to price-related heterogeneity, since fuel and distance affect price directly, but $\sigma_\gamma$ governs preference for the ULCC characteristic, which did not touch price or cost anywhere in the data-generating process as originally built. The fare-generation code was revised to add a genuine channel from ULCC rival count to price specifically (the same competitive discount mechanism described in Section C.2), and the corresponding instrument was added once that channel existed to identify against. Even with this addition, $\sigma_\gamma$ remains weakly identified, as documented in the Estimation Results section; the addition made $\sigma_\gamma$ identification possible in principle, not strong in practice.

## Technical Appendix D: GMM Standard Error Derivation

This appendix derives the standard error formula used for the 5 linear coefficients in Table 3, shows the algebraic simplification behind it step by step, and documents the p-value methodology and the scope limitation already disclosed in the Limitations section.

## D.1 The Standard Asymptotic Variance Formula

For 2-step efficient GMM, the standard asymptotic variance of the estimated coefficients is

$$\text{Avar}(\hat{\beta}) = \frac{1}{N}\left[G'S^{-1}G\right]^{-1}$$

where $G$ is the Jacobian of the moment condition with respect to $\beta$, and $S$ is the moment condition's covariance. In this analysis's notation, the moment condition is $g_i = Z_i\xi_i(\beta)$, and since $\xi_i(\beta) = \delta_i - X_i'\beta$, the Jacobian is

$$G = E\left[\frac{\partial g_i}{\partial \beta}\right] = -E[Z_iX_i'], \qquad \hat{G} = -\frac{1}{N}Z'X$$

$S$, the moment covariance, is $E[g_ig_i'] = E[\xi_i^2Z_iZ_i']$. This is exactly the quantity `compute_efficient_weight_matrix` already computes, before inverting it into the second-step weight matrix $W$; that is, $W = \hat{S}^{-1}$ already, no separate computation is required.

## D.2 Algebraic Simplification

Substituting $\hat{G} = -(1/N)Z'X$ and $S^{-1} = W$ into the standard formula:

$$G'S^{-1}G = \left[-\frac{1}{N}X'Z\right]W\left[-\frac{1}{N}Z'X\right] = \frac{1}{N^2}\left(X'ZWZ'X\right)$$

$$\text{Avar}(\hat{\beta}) = \frac{1}{N}\left[\frac{1}{N^2}\left(X'ZWZ'X\right)\right]^{-1} = \frac{1}{N}\times N^2\left(X'ZWZ'X\right)^{-1} = N\left(X'ZWZ'X\right)^{-1}$$

so the final, implemented formula is

$$\mathrm{Var}(\hat{\beta}) = N\left(X'ZWZ'X\right)^{-1}$$

Standard errors are the square roots of this matrix's diagonal. Every quantity on the right-hand side ($X$, the design matrix; $Z$, the instrument matrix; $W$, the second-step weight matrix; $N$, the number of observations) is already computed inside the 2-step efficient GMM procedure, so this required no new estimation machinery beyond assembling pieces already on hand.

## D.3 Scope Limitation

This formula, and the standard errors reported in Table 3, cover only the 5 linear coefficients ($\alpha$, $\beta_{\text{seat pitch}}$, $\beta_{\text{nonstop}}$, $\beta_{\text{ULCC}}$, $\beta_{\text{JetBlue}}$), treating the final nonlinear parameter estimates ($\hat{\sigma}_\alpha$, $\hat{\sigma}_\gamma$) as fixed and known, not as additional sources of sampling uncertainty. A fully correct joint standard error spanning all 7 parameters together would require numerically differentiating the moment conditions with respect to the nonlinear parameters as well, a substantially larger undertaking. Mean utility ($\delta$) is only implicitly defined by the inner loop's contraction mapping, so obtaining the full Jacobian would require differentiating the moment conditions through that inner solution, either analytically via the implicit function theorem or numerically. This was treated as out of scope for this analysis. The repeated-run stability evidence documented in the Estimation Results and Limitations sections, $\hat{\sigma}_\alpha$ agreeing closely across independent outer loop searches while $\hat{\sigma}_\gamma$ does not, is the available diagnostic evidence on the 2 nonlinear parameters' relative reliability in place of a formal standard error.

## D.4 P-Values and Significance

P-values reported alongside the standard errors use a 2-tailed test against the standard normal distribution, $p = 2\left(1 - \Phi(|t|)\right)$, following GMM's large-sample asymptotic theory. With 1,560 observations, the analysis uses the normal approximation rather than a finite-sample t-distribution.

# Technical Appendix E: Gamma Identification Grid-Search Diagnostics

This appendix documents a grid search over the 2 nonlinear parameters, computing the GMM objective at each point on a fixed grid rather than relying only on where a local numerical search happens to converge. This shows directly, rather than infers, how strongly the GMM objective responds to $\sigma_\alpha$ and $\sigma_\gamma$ over the tested parameter range.

## E.1 Method

The outer loop's Nelder-Mead search (described in the Empirical Strategy section) is a local optimizer: it reports where it converges, not the shape of the objective function around that point. A grid search instead computes the full GMM objective at every combination of $\sigma_\alpha$ and $\sigma_\gamma$ on a fixed grid, showing whether the objective has a genuine, well-defined minimum in each parameter or not. The grid used here spans $\sigma_\alpha \in \{0.002, 0.003, 0.004, 0.005, 0.006\}$ and $\sigma_\gamma \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2\}$, 35 combinations in total, each requiring a full inner-loop inversion across all 492 markets and a full 2-step efficient GMM computation, using the same code and the same 4 instruments as the main estimation.

## E.2 The Global Minimum

*Table 15: GMM Objective, Full Grid*

| $\sigma_\alpha \setminus \sigma_\gamma$ | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 |
|---|---|---|---|---|---|---|---|
| 0.002 | 0.891 | 0.983 | 1.218 | 1.622 | 2.217 | 3.003 | 3.961 |
| 0.003 | 0.712 | 0.823 | 1.096 | 1.573 | 2.287 | 3.242 | 4.407 |
| 0.004 | **0.651** | 0.786 | 1.103 | 1.667 | 2.529 | 3.697 | 5.129 |
| 0.005 | 0.842 | 1.006 | 1.373 | 2.032 | 3.057 | 4.464 | 6.202 |

| $\sigma_\alpha \setminus \sigma_\gamma$ | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 |
|---|---|---|---|---|---|---|---|
| 0.006 | 1.396 | 1.596 | 2.018 | 2.773 | 3.959 | 5.607 | 7.658 |

The global minimum across all 35 grid points is at $\sigma_\alpha = 0.004$, $\sigma_\gamma = 0.0$, an objective value of 0.651. This is not a local search artifact: it is the smallest value the GMM objective takes anywhere on this grid. $\sigma_\alpha = 0.004$ is the true value exactly; $\sigma_\gamma = 0.0$ is the lower boundary of the tested range, far from the true value of 1.0.

## E.3 The Sigma_Alpha Profile: a Genuine Interior Minimum

Holding $\sigma_\gamma$ fixed at 0 and varying $\sigma_\alpha$ across the grid:

*Table 16: GMM Objective vs. $\sigma_\alpha$ (at $\sigma_\gamma = 0$)*

| $\sigma_\alpha$ | 0.002 | 0.003 | 0.004 | 0.005 | 0.006 |
|---|---|---|---|---|---|
| Objective | 0.891 | 0.712 | 0.651 | 0.842 | 1.396 |

The objective falls, reaches a minimum exactly at $\sigma_\alpha = 0.004$, the true value, and rises again on both sides. This U-shape, with a clear interior minimum and the objective increasing in both directions away from it, provides direct evidence that the data strongly identify $\sigma_\alpha$ within the range examined: moving $\sigma_\alpha$ away from its true value in either direction makes the model fit the instruments' moment conditions worse.

## E.4 The Sigma_Gamma Profile: No Interior Minimum

Holding $\sigma_\alpha$ fixed at 0.004 (its near-true value) and varying $\sigma_\gamma$ across the grid:

*Table 17: GMM Objective vs. $\sigma_\gamma$ (at $\sigma_\alpha = 0.004$)*

| $\sigma_\gamma$ | 0.0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2 |
|---|---|---|---|---|---|---|---|
| Objective | 0.651 | 0.786 | 1.103 | 1.667 | 2.529 | 3.697 | 5.129 |

Unlike the $\sigma_\alpha$ profile, this is monotonically increasing across the entire tested range, including at $\sigma_\gamma = 1.0$, the true value, where the objective (3.697) is more than 5 times larger than at $\sigma_\gamma = 0$. There is no interior minimum anywhere in this range: the objective simply prefers $\sigma_\gamma$ to be as small as possible. This is a direct, visual confirmation of the identification argument made in the Estimation Results and Empirical Strategy sections: the available instruments provide the price variation needed to pin down $\sigma_\alpha$, but the instrument set provides no excluded variation that separates the effect of the ULCC characteristic from price, so the GMM objective has no genuine incentive to move $\sigma_\gamma$ away from 0 toward its true value.

### E.5 Consistency With the Outer Loop's Result

This grid search was run entirely independently of the actual outer loop search reported in the Estimation Results section, using the same underlying code but a different search procedure (a fixed grid rather than Nelder-Mead). That the grid's minimum lands at $\sigma_\alpha \approx 0.004$, matching the outer loop's converged estimate of 0.003813, and at $\sigma_\gamma = 0$, matching the outer loop's converged estimate of 0.000003, confirms the outer loop did not converge to a poor local optimum or fail to search properly: it found the same region of the objective surface identified by the independent grid search, whose minimum over the tested grid occurs at $\sigma_\gamma = 0$ and whose objective increases steadily as $\sigma_\gamma$ moves toward and beyond its true value.

## Technical Appendix F: Solver Integrity Audit

This appendix documents a solver-integrity verification added during the final audit of the merger simulation, and the check's result across every equilibrium price underlying Table 9.

## F.1 The Verification Added

`solve_equilibrium_prices`, the function underlying every equilibrium price in Table 9, calls `scipy.optimize.fsolve` to find the price vector satisfying the first-order pricing conditions. A numerical solver can occasionally report a result without having genuinely converged, so trusting its output unconditionally risks silently propagating an incorrect price vector into every downstream result. `solve_equilibrium_prices` therefore checks convergence directly, rather than assuming it.

## F.2 The Check

`solve_equilibrium_prices` verifies 2 things independently before accepting a solution, using `fsolve`'s `full_output=True` option:

1. `fsolve`'s own internal convergence flag (`ier == 1`), which reflects whether the solver itself considers the solve successful.
2. The first-order condition residuals, independently recomputed at the returned price vector rather than inferred from `fsolve`'s internal state, checked against an absolute tolerance: $\max_j |FOC_j(p)| < 10^{-6}$.

These 2 checks are distinct: the first reflects `fsolve`'s own assessment of its solve, the second independently verifies the actual returned solution against the equations it was supposed to satisfy. If either check fails, the function raises a `RuntimeError` rather than silently returning a questionable price vector, the same fail-loudly convention already used for non-convergence in the inner loop's contraction mapping (described in the Empirical Strategy section). This check is permanent, built directly into the function itself: every future call to `solve_equilibrium_prices` is automatically audited.

### F.3 The Audit Result

Every equilibrium price underlying Table 9, the pre-merger validation solve and the post-merger counterfactual solve, on all 3 anchor routes and all 12 quarters each, was checked under this verification (72 solves in total: 3 routes $\times$ 12 quarters $\times$ 2 ownership scenarios). This was independently re-run while preparing this appendix, using the same code and the same final estimated parameters, and all 72 of 72 solves converged, with 0 failures.

### F.4 What This Establishes, and What It Does Not

This audit establishes that every price behind Table 9 is a converged solution to the first-order conditions it was supposed to solve, not an unverified numerical result. It does not establish that the underlying economic model or estimated parameters are correct; that is a separate question, addressed throughout the Estimation Results and Limitations sections. A converged solution to the wrong first-order conditions would still pass this audit. What this audit rules out specifically is numerical failure: a solver silently returning a price vector that does not actually solve the equations it was asked to solve.

## Technical Appendix G: Repository Structure and Reproducibility

This appendix documents the code repository's structure, the fixed random seeds behind every result in this analysis, and the steps needed to regenerate the panel and reproduce the estimation from scratch.

### G.1 Repository Structure

| Directory | Contents |
|---|---|
| src/simulation/ | The 6 modules building the data-generating process: market structure, time structure, carrier products and price generation, demand primitives, the real fuel and fare data modules, and the calibration check. |
| src/estimation/ | The BLP estimation pipeline: instrument construction, the inner loop (plain and SQUAREM-accelerated), GMM, the outer loop, and elasticities. |
| src/effects/ | The merger simulation: marginal cost recovery, ownership matrix construction, the equilibrium price solver, and the consumer welfare calculation. |
| notebooks/ | A numbered notebook for each major build or estimation stage (01 through 13), demonstrating and validating each piece of src/ in turn. |
| data/raw/ | Real source data and anchor figures extracted from the case record and public data sources. |
| data/processed/ | The fixed, generated panel (simulated_panel.csv), the true parameters (true_parameters.json), and the final estimation result (final_estimation_result.json). |

## G.2 Random Seeds

Every stage of construction and estimation that involves randomness uses a fixed seed, so the entire analysis, from the constructed panel through to the final estimates, is exactly reproducible from the same code.

| Stage | Seed | Where Used |
|---|---|---|
| Route panel construction | 42 | `market_structure.build_route_panel` |
| Route-quarter panel construction (exit events) | 99 | `time_structure.build_route_quarter_p` |
| Carrier product panel (fares, $\xi$) | 7 | `carrier_products.build_carrier_produ` |
| Demand primitives (consumer draws, shares) | 123 | `demand_primitives` |
| Fixed simulation draws for estimation ($\nu^{\alpha}$, $\nu^{\gamma}$) | 2024 | `inner_loop.draw_fixed_simulation_dra` called with `n_draws=100` throughout every notebook and script in `src/estimation/` and `src/effects/` |

The first 4 seeds are recorded directly in `true_parameters.json`. The fifth is not a default value inside the function itself; it is a convention used consistently by every caller throughout the codebase, and every notebook and script in this analysis passes it explicitly.

## G.3 Reproducing the Fixed Dataset

The panel used throughout this analysis (`simulated_panel.csv`) is generated once, with the seeds above, and then held fixed for every subsequent notebook and script, rather than regenerated on the fly each time. This is deliberate: since the outer loop search (Empirical Strategy section)

and the merger simulation (Spirit Exit and Merger Simulation section) are both computationally expensive, holding the underlying data fixed lets every later step build on the same panel without introducing new randomness at each stage. `notebooks/07_generate_final_dataset.ipynb` runs functions 1 through 4 of `src/simulation/` in sequence with the seeds above, saves the resulting panel to `data/processed/simulated_panel.csv`, and saves the true parameters (used only for the oracle comparisons throughout this analysis, never by the estimation itself) to `data/processed/true_parameters.json`. Re-running this notebook regenerates an identical panel, since every source of randomness in it is seeded.

## G.4 Reproducing the Estimation

`notebooks/10_gmm_and_outer_loop.ipynb` runs the full nested fixed point search described in the Empirical Strategy section and saves the final estimated parameters to `data/processed/final_estimation_result.json`. The outer loop's Nelder-Mead search is itself deterministic, given a fixed starting point, a fixed objective function, and the fixed simulation draws from Section G.2. So re-running this notebook on the same panel and the same code reproduces the same estimated parameters to the reported precision. This was directly confirmed during the preparation of this analysis, when the search was re-run independently on 2 separate occasions and converged to the same result (0.003813, 0.000003) both times. Every subsequent notebook (elasticities, the Spirit exit and merger simulation, and the appendices) reads `final_estimation_result.json` directly rather than re-running the search, so the downstream analysis uses the same saved estimates rather than introducing additional estimation variation.